PSDC
Physical Design Training

BITCOIN PHYSICAL DESIGN PROJECT

Group 1 Presentation

2 November 2026



PSDC Physical Design Training

PROJECT ORGANISATION

MEET THE TEAM

Roles and ownership across the Bitcoin implementation sprint

SCRUM MASTER

ASYRAF

Formality verification & RTL ECO



HAMZAH

Timing closure & manual ECO



SAM

Timing analysis & sweep planning



HAZIM

Scripts & FC / PT correlation



ANIS

Frequency & recipe optimisation



SYAMIM

Project setup, sweep launch & reporting



DESIGN BACKGROUND

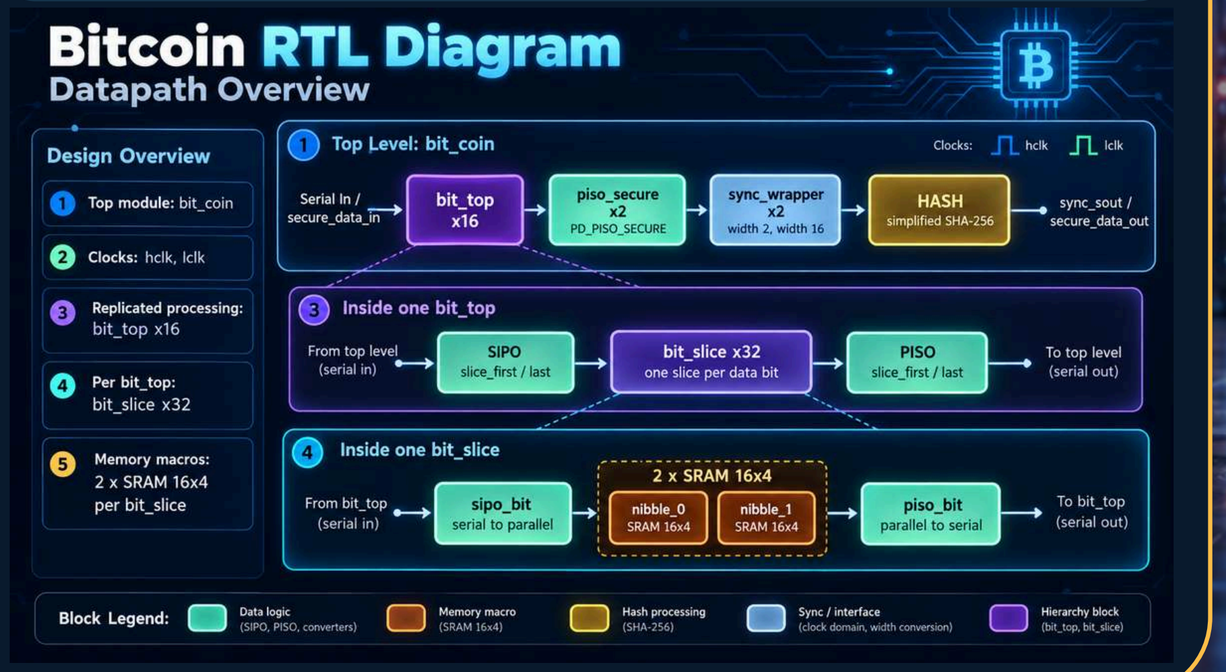
WHAT ARE WE IMPLEMENTING?

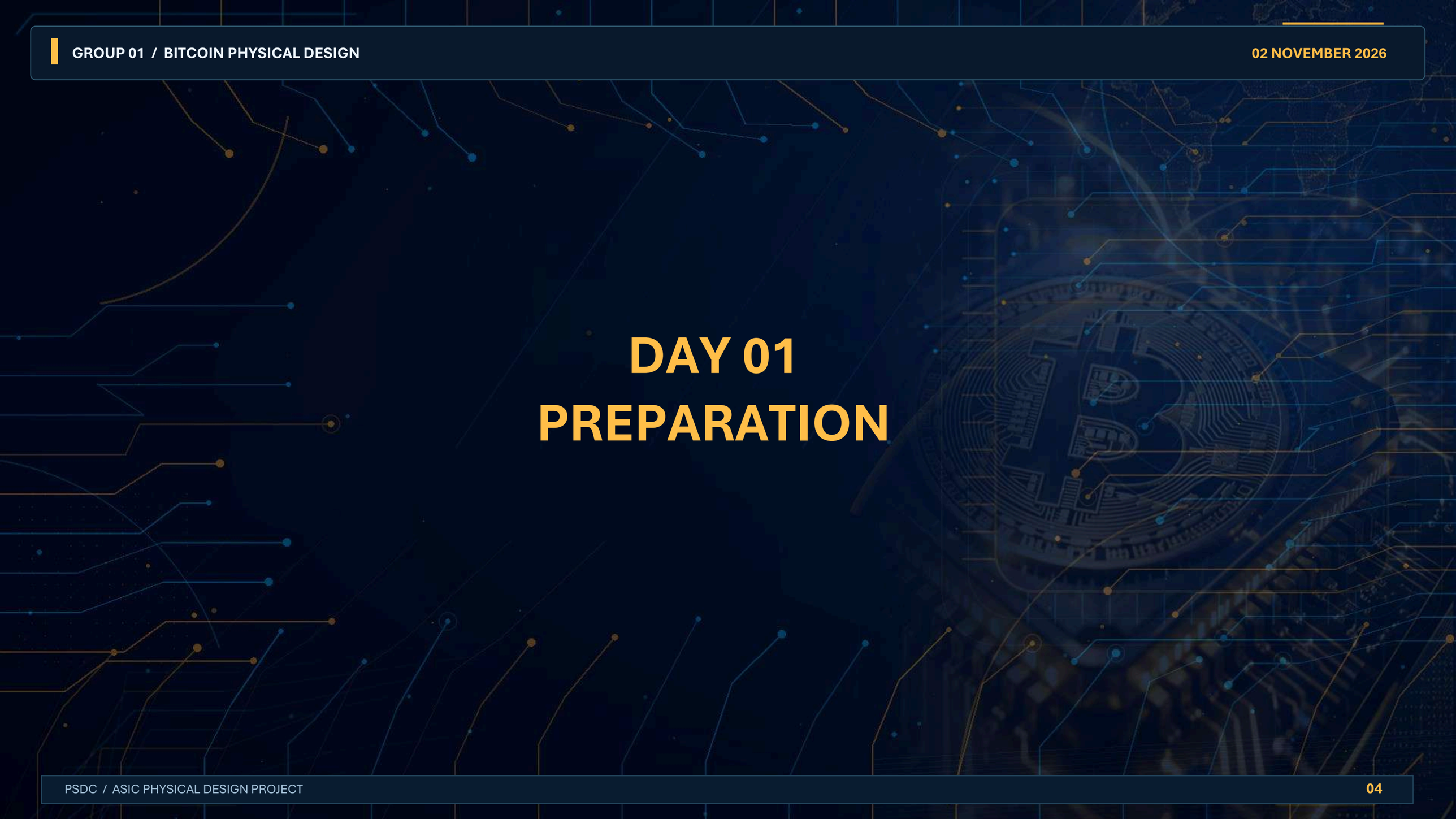
Bitcoin-inspired digital architecture for a mining-oriented processing datapath

PROJECT CONTEXT

- **Mining concept**
  Parallel hashing accelerates computation across repeated datapath structures.
- **Digital core**
  Design includes serial-to-parallel data conversion, SRAM storage and a simplified SHA-256 block.
- **Our scope**
  Physical implementation, timing / PPA optimisation and functional equivalence verification.

Bitcoin Datapath Overview





DAY 01
PREPARATION

DAY 01 • PREPARATION

INITIAL SETUP & DEBUGGING

Flow preparation, input collateral, init_design and complete baseline run

ACTIVITIES

- **Understand the flow**
  Makefile, design_setup.tcl and fc_setup.tcl
- **Fix start-up errors**
  Validate init_design and solve missing inputs or scripting problems.
- **Launch baseline**
  Complete the 1 GHz reference implementation.

Fixes on design_setup.tcl :

Incorrect library path

```
logs_fc/init_design.log:1534:Error:  Unable to open file `bit_coin.v': in search_path {. /home/synopsys/fusioncompiler/W-2024.09-SP3/dw/syn_ver /home/synopsys/fusioncompiler/W-2024.09-SP3/dw/sim_ver /home/synopsys/fusioncompiler/W-2024.09-SP3/libraries/syn ./rm_user_plugin_scripts ./rm_tech_scripts ./rm_fc_scripts ./rm_setup ./examples . data data/rtl scripts data/lib/sram/ndms data/lib/stdcell_hvt/ndms data/lib/stdcell_rvt/ndms data/lib/stdcell_lvt/ndms data/lib/sram/lib_dbs data/lib/stdcell_hvt/db_nldm_gm data/lib/stdcell_lvt/db_nldm data/lib/stdcell_rvt/db_nldm data/tech/icv_drc}. (VER-41)
```

```
logs_fc/init_design.log:223:Error: no files matched glob patterns "data/lib/dbs_nwell/*.db data/lib/sram/lib_dbs/*.db"
logs_fc/init_design.log:1131:          puts "RM-error: env(RM_VARFILE) specified but not found"
logs_fc/init_design.log:1207:                                        puts "RM-error: Library does not exist for ${BLOCK
}${LIBRARY_SUFFIX}. Exiting"
```

Solution :

```
set SUPPLEMENTAL_SEARCH_PATH          "[list . data \
data/rtls scripts \
data/libs/sram/ndms \
data/libs/stdcell_hvt/ndms \
data/libs/stdcell_rvt/ndms \
data/libs/stdcell_lvt/ndms \
data/libs/sram/lib_dbs \
data/libs/stdcell_hvt/db_nldm_gm \
data/libs/stdcell_lvt/db_nldm \
data/libs/stdcell_rvt/db_nldm \
data/tech/icv_drc]" ;
```

```
214 set LINK_LIBRARY   [glob data/libs/dbs_nwell/*.db \
215                     data/libs/sram/lib_dbs/*.db]
```

DAY 01 • PREPARATION

INITIAL SETUP & DEBUGGING

Flow preparation, input collateral, init_design and complete baseline run

Incorrect UPF file path

Invalid SDC syntax

Fixes on data/ upf/ snps_bit_coin_flat.upf :

```
logs_fc/init_design.log:8521:Tool Error: File '/home/hyong/bitcoin/bitcoin_pnr_lab/data/upf/re
fine_bc_flat.upf' cannot be found using search_path of: '. /home/synopsys/fusioncompiler/W-202
4.09-SP3/dw/syn_ver /home/synopsys/fusioncompiler/W-2024.09-SP3/dw/sim_ver /home/synopsys/fusi
oncompiler/W-2024.09-SP3/libraries/syn ./rm_user_plugin_scripts ./rm_tech_scripts ./rm_fc_scri
pts ./rm_setup ./examples . data data/rtls scripts data/libs/sram/ndms data/libs/stdcell_hvt/n
dms data/libs/stdcell_rvt/ndms data/libs/stdcell_lvt/ndms data/libs/sram/lib_dbs data/libs/std
cell_hvt/db_nldm_gm data/libs/stdcell_lvt/db_nldm data/libs/stdcell_rvt/db_nldm data/tech/icv_
drc'. (FILE-002) issued on line 1863
logs_fc/init_design.log:8522:Tool Error: problem in load_upf issued on line 1864
logs_fc/init_design.log:8523:Tool Error: Nothing matched for supply_port_name (SEL-005) issued
 on line 1876
logs_fc/init_design.log:8524:Tool Error: bad value specified for option supply_port_name issue
d on line 1877
logs_fc/init_design.log:8525:Tool Error: Nothing matched for supply_port_name (SEL-005) issued
 on line 1881
logs_fc/init_design.log:8526:Tool Error: bad value specified for option supply_port_name issue
d on line 1882
logs_fc/init_design.log:8527:Tool Error: Nothing matched for supply_port_name (SEL-005) issued
 on line 1886
logs_fc/init_design.log:8528:Tool Error: bad value specified for option supply_port_name issue
d on line 1887
logs_fc/init_design.log:8529:Tool Error: Nothing matched for supply_port_name (SEL-005) issued
```

Solution :

```
60 # set UPF_DIR upf
61 load_upf data/upf/refine_bc_flat.upf
62 add_power_state -supply SS -state SS_ON {-supply_expr {power == {FULL_ON 0.85} }}
63 add_power_state -supply SS -state GND   {-supply_expr {ground == `{FULL_ON,0.0}}}
64 add_power_state -supply SSL -state SSL_ON {-supply_expr {power == {FULL_ON 0.78} }}
```

Fixes on data/ sdc / bit_coin.sdc :

```
logs_fc/init_design.log:2302:Error: unknown command 'set_loads' (CMD-005)
logs_fc/init_design.log:2304:Error: Errors reading SDC file:
logs_fc/init_design.log:2340:Error: unknown command 'set_loads' (CMD-005)
logs_fc/init_design.log:2342:Error: Errors reading SDC file:
logs_fc/init_design.log:2379:Error: unknown command 'set_loads' (CMD-005)
logs_fc/init_design.log:2381:Error: Errors reading SDC file:
logs_fc/init_design.log:2551:           puts "RM-error: CTS_NDR_RULE_NAME($
ted yet. Can not associate it with root nets!"
logs_fc/init_design.log:2601:           puts "RM-error: CTS_INTERNAL_NDR_RU
) hasn't been created yet. Can not associate it with internal nets!"
```

Solution :

```
 67
 68 set_load 0.2 [all_outputs ]
~
~
```

DAY 01 • PREPARATION

INITIAL SETUP & DEBUGGING

Flow preparation, input collateral, init_design and complete baseline run

Fixes on data/ rtls :

```
logs_fc/init_design.log:2134:Error: cant find lib bit_coin.nlib
logs_fc/init_design.log:2156:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2160:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2164:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2167:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2170:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2183:Error: cant find lib bit_coin.nlib
logs_fc/init_design.log:2203:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2207:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2211:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2214:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2217:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2231:Error: cant find lib bit_coin.nlib
logs_fc/init_design.log:2251:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2255:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2259:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2262:Error: Nothing matched for -object_list (SEL-005)
logs_fc/init_design.log:2265:Error: Nothing matched for -object_list (SEL-005)
```

Invalid rtl syntax

Solution :

```
13 //
14 //  FILE    : files_bit_coin.f
15 //
16 //  VERSION : 1.0
17 //
18 ////////////////////////////////////
19
20 data/rtls/bit_coin.v
21 -y data/rtls
22 +libext+.v
```

files_bit_coin.f

```
14 //  FILE    : files_bit_coin.f
15 //
16 //  VERSION : 1.0
17 //
18 ////////////////////////////////////
19
20 ../../data/rtls/bit_coin.v
21 -y ../../data/rtls
22 +libext+.v
```

files_bit_coin.f_orig

```
14 //  FILE    : files_bit_top.f
15 //
16 //  VERSION : 1.0
17 //
18 ////////////////////////////////////
19
20 rtls/bit_top.v
21 -y rtls
22 +libext+.v
```

files_bit_slice.f

DAY 01 • PREPARATION

DEBUGGING ERROR OCCUR AT CTS

Fixes on rm_user_plugin_scripts / clock_opt_cts_pre_script.tcl :

```
logs_fc/clock_opt_cts.log:1126:         puts "RM-error: env(RM_VARFILE) specified but not found"
logs_fc/clock_opt_cts.log:4129:Error: can't read "pins_bs_0_20": no such variable
logs_fc/clock_opt_cts.log:4132:Error: can't read "pins_bs_1_20": no such variable
logs_fc/clock_opt_cts.log:4135:Error: can't read "pins_bs_2_20": no such variable
logs_fc/clock_opt_cts.log:4138:Error: can't read "pins_bs_3_20": no such variable
logs_fc/clock_opt_cts.log:4141:Error: can't read "pins_bs_4_20": no such variable
logs_fc/clock_opt_cts.log:4144:Error: can't read "pins_bs_5_20": no such variable
logs_fc/clock_opt_cts.log:4147:Error: can't read "pins_bs_6_20": no such variable
logs_fc/clock_opt_cts.log:4150:Error: can't read "pins_bs_7_20": no such variable
logs_fc/clock_opt_cts.log:4153:Error: can't read "pins_bs_8_20": no such variable
logs_fc/clock_opt_cts.log:4156:Error: can't read "pins_bs_9_20": no such variable
logs_fc/clock_opt_cts.log:4159:Error: can't read "pins_bs_10_20": no such variable
logs_fc/clock_opt_cts.log:4162:Error: can't read "pins_bs_11_20": no such variable
logs_fc/clock_opt_cts.log:4165:Error: can't read "pins_bs_12_20": no such variable
logs_fc/clock_opt_cts.log:4168:Error: can't read "pins_bs_13_20": no such variable
logs_fc/clock_opt_cts.log:4171:Error: can't read "pins_bs_14_20": no such variable
logs_fc/clock_opt_cts.log:4174:Error: can't read "pins_bs_15_20": no such variable
logs_fc/clock_opt_cts.log:4212:Error: Nothing matched for -sinks (SEL-005)
logs_fc/clock_opt_cts.log:4213:Error: No physical net to buffer found. (CTP-048)
logs_fc/clock_opt_cts.log:4216:Error: Nothing matched for -sinks (SEL-005)
```

Variable mismatch

Missing $ symbol

Solution :

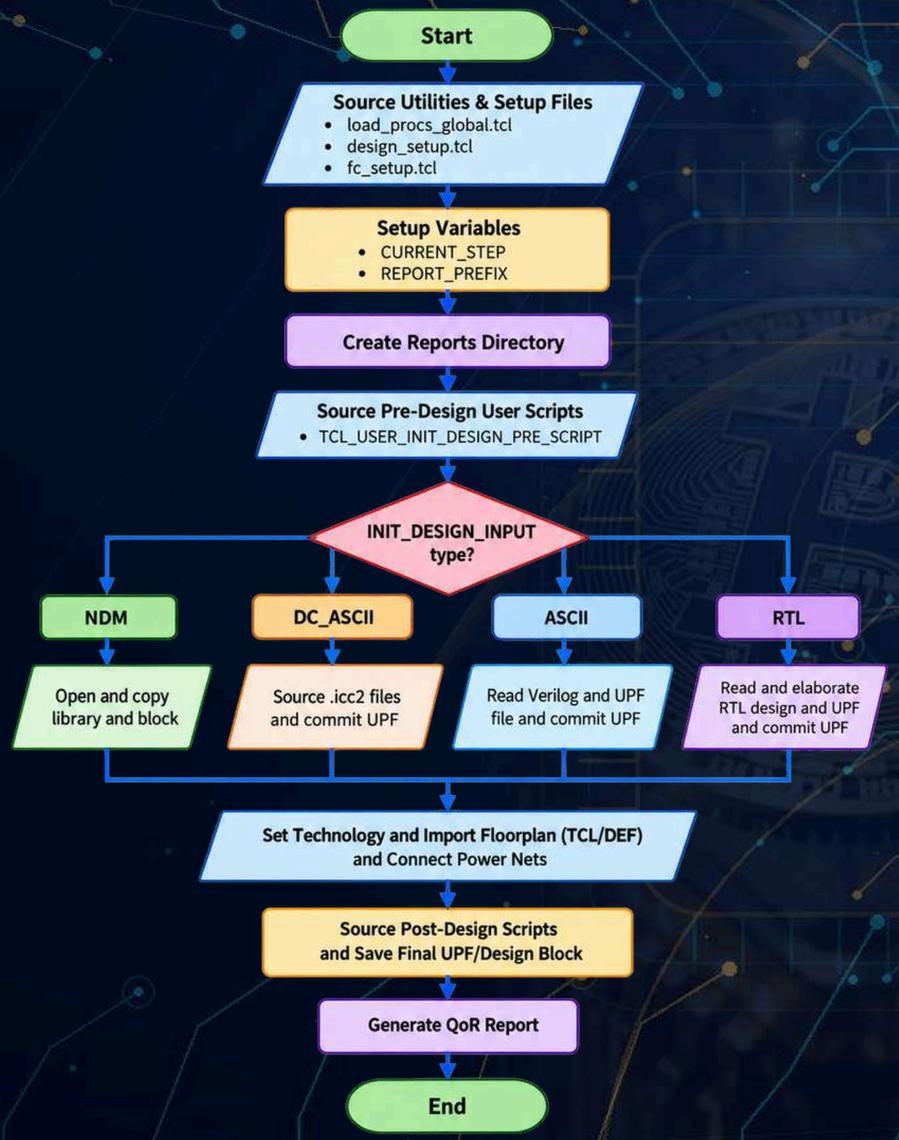
```
33 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_0_21]
34 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_1_21]
35 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_2_21]
36 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_3_21]
37 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_4_21]
38 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_5_21]
39 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_6_21]
40 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_7_21]
41 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_8_21]
42 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_9_21]
43 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_10_21]
44 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_11_21]
45 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_12_21]
46 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_13_21]
47 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_14_21]
48 create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_15_21]
```

```
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_0_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_1_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_2_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_3_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_4_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_5_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_6_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_7_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_8_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_9_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_10_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_11_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_12_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_13_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_14_31]
create_clock_buffer -clock lclk -sinks [get_pins $pins_bs_15_31]
```

DAY 02
BACKGROUND STUDY & KEY DESIGN INFO COLLECTION

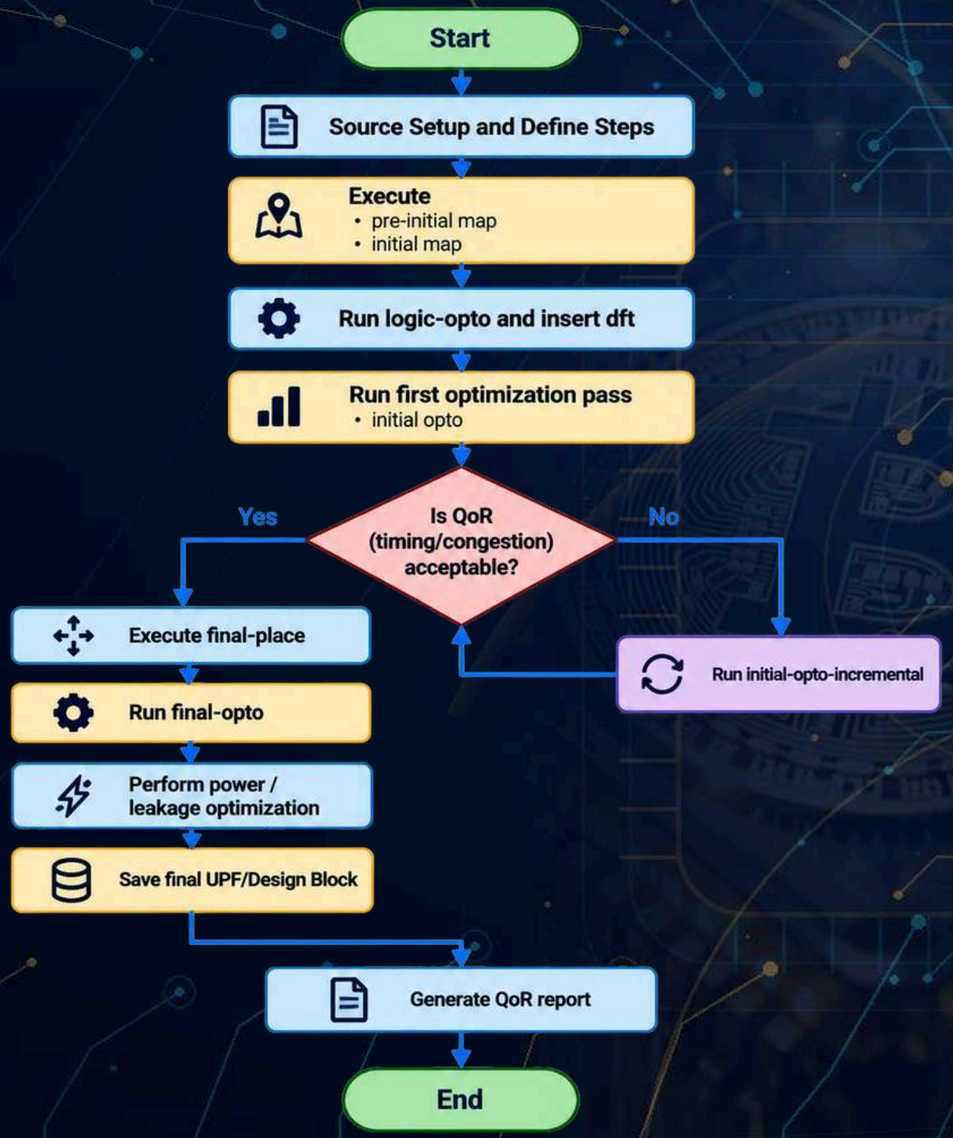
DAY 02 • BACKGROUND STUDY & KEY DESIGN INFO COLLECTION

RM FLOW - INIT_DESIGN



DAY 02 • BACKGROUND STUDY & KEY DESIGN INFO COLLECTION

RM FLOW -COMPILE



DAY 02 • BACKGROUND STUDY & KEY DESIGN INFO COLLECTION

BITCOIN DESIGN OVERVIEW

BIT_TOP BLOCKS

16

top-level instances

BIT_SLICE BLOCKS

512

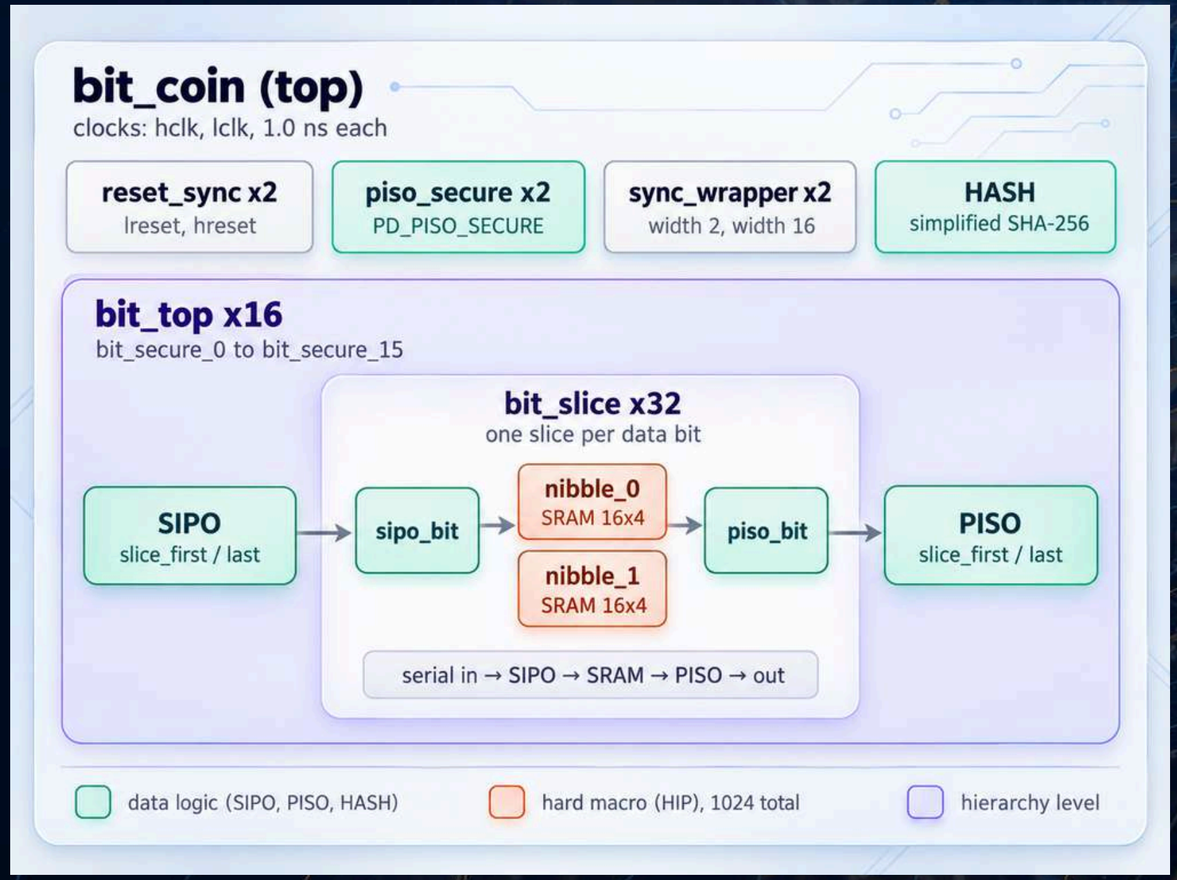
16 × 32

SRAM MACROS

1,024

16 × 32 × 2

Bitcoin Block Diagram



Control & data conversion: SIPO / PISO • Hash computation: simplified SHA-256 • Memory: 16×4 SRAM

DAY 02 • BACKGROUND STUDY & KEY DESIGN INFO COLLECTION

BITCOIN RTL HIERARCHY & DATA FLOW

Hierarchical replication drives 1,024 SRAM macro instances

BIT_COIN : TOP-LEVEL DATAPATH

bit_top ×16
bit_secure_0 ... 15

→ **PISO ×2**
secure output

→ **SYNC**
wrapper

→ **HASH**
SHA-256

→ **sync_sout**
2-bit

INSIDE ONE BIT_TOP

SIPO
first / last

→ **bit_slice ×32**
one per data bit

→ **PISO**
first / last

INSIDE ONE BIT_SLICE

sipo_bit
serial → parallel

→ **2 × SRAM 16×4**
nibble_0 / nibble_1

→ **piso_bit**
parallel → serial

TOTAL SRAM MACROS

1,024

16 × 32 × 2

DAY 02 • BACKGROUND STUDY & KEY DESIGN INFO COLLECTION

KEY DESIGN INFORMATION

Confirm the measured design characteristics from your actual reports

CLOCKS

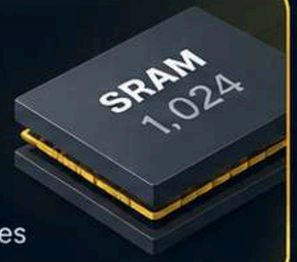
2

hclk, lclk

SRAM MACROS

1,024

512 slices × 2 memories



POWER DOMAINS

3

PD_PISO_SECURE, PD_SIPO_SECURE, TOP



GATE COUNT

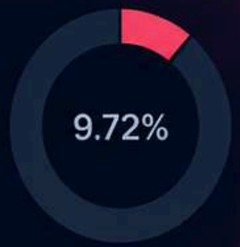
193,362

at Chip_Finish

UTILIZATION

9.72%

at Chip_Finish



HIGH FANOUT NETS

retention_signals[1]	9282
retention_signals[0]	28832
data_valid	13634

DAY 02 • BACKGROUND STUDY & KEY DESIGN INFO COLLECTION

SUB-BLOCKS

Control vs Data blocks , HIP presence, key clocks and power domains

Data block (including storage/compute) | Control block (glue / control logic) | Mixed / wrapper (top-level or mixed function) | HIP present | No HIP | hclk High-speed clock | lclk Low-speed clock

Sub-block	Category	Key Clock(s)	Power Domain	HIP	Notes
bit_coin (top)	Mixed / wrapper	hclk lclk	TOP (0.85 V)	Contains all 1024 SRAMs	Top-level integration wrapper
bit_top ×16	Data block (wrapper)	lclk	TOP; edge SIPO/PISO sit in switchable domains	Contains SRAMs	bit_secure_0 to bit_secure_15
bit_slice ×32	Data block (wrapper)	lclk	TOP; sipo_bit and piso_bit inside are switchable	Contains 2 SRAMs	One slice per data bit
SIPO	Data block	lclk	PD_SIPO_SECURE (0.78 V, switchable)	No	Shift register, serial to parallel
PISO	Data block	lclk	PD_PISO_SECURE (0.78 V, switchable)	No	Shift register, parallel to serial
SRAM2RW16x4	Data block (storage)	lclk	TOP	Yes, the only HIP macro	Hard macro memory block
HASH	Data / compute datapath	hclk	TOP	No, if plain logic	Compute datapath with some control
reset_sync ×2	Control block	hclk lclk	TOP	No	Reset synchronization / glue logic
sync_wrapper ×2	Control block	hclk lclk	TOP	No	Synchronization wrapper / glue logic

Two main clocks: **hclk and lclk**

Switchable secure domains: **PD_SIPO_SECURE and PD_PISO_SECURE**

Only hard IP macro: **SRAM2RW16x4,** replicated to 1024 instances overall

DAY 02 • ANALYSIS

1G RUN (INIT_DESIGN TO ICV_IN_DESIGN) TABLE

STAGES	RUN TIME (HOUR)	UTILIZATION %	GATE / CELL COUNT
INIT_DESIGN	0.16	N/A	N/A
COMPILE	1.24	9.53	196,525
CLOCK_OPT_CTS	1.41	9.75	203,731
CLOCK_OPT_OPTO	0.69	9.62	197,593
ROUTE_AUTO	0.27	9.62	197,593
ROUTE_OPT	0.92	9.72	193,362
CHIP_FINISH	0.39	9.72	193,362
ICV_IN_DESIGN	0.32	9.72	193,362

DAY 02 • ANALYSIS

1G RUN (INIT_DESIGN TO ICV_IN_DESIGN) FORMALITY CHECK

```
For further details on Synopsys Auto Setup Mode: Type man synopsys_auto_setup
****************************************************************************************

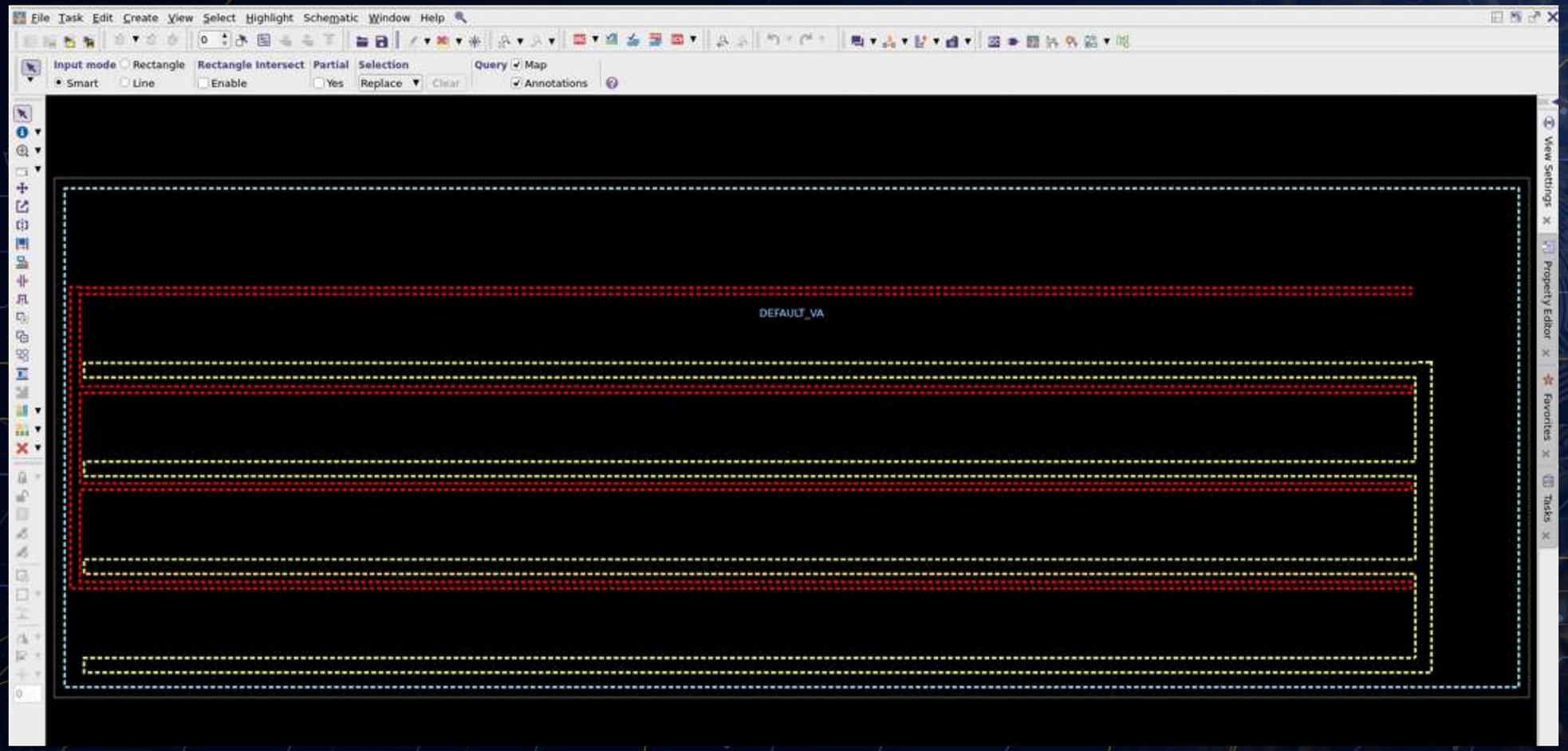

************************************ Verification Results ************************************
Verification SUCCEEDED
   ATTENTION: Verification was run with all UPF supplies constrained to their ON state.
              This is only a partial verification result as it does not cover all
              operational states.
   ATTENTION: synopsys_auto_setup mode was enabled.
              See Synopsys Auto Setup Summary for details.
----------------------------------------------------------------------------------------
 Reference design: r:/WORK/bit_coin
 Implementation design: i:/WORK/bit_coin
 119990 Passing compare points
----------------------------------------------------------------------------------------------------
Matched Compare Points     BBPin    Loop   BBNet     Cut     Port   PDCut     DFF      LAT   TOTAL
----------------------------------------------------------------------------------------------------
Passing (equivalent)       28672       0       0    1025      121   10946   79226        0  119990
Failing (not equivalent)       0       0       0       0        0       0       0        0       0
Not Compared
  Don't verify                 0       0       0       0        4       0       0        0       4
****************************************************************************************************
1
save_session -replace ./fm_sess/fm_sess_${DESIGN_TOP}_rtl2gate.fss
Error: No session file generated. Reason : Parent directory does not exist. (FM-427)
0
redirect -file ${REPORTS_DIR}/${DESIGN_TOP}.fm.rtl2gate.rpt {report_status}
exit

Maximum memory usage for this session: 2900 MB
CPU usage for this session: 874.86 seconds ( 0.24 hours )
Current time: Fri Oct 23 14:50:46 2026
Elapsed time: 879 seconds ( 0.24 hours )

Thank you for using Formality (R)!
[pduser11@psdc-eda03 bitcoin_pnr_lab]$
```

DAY 02 • ANALYSIS

POWER DOMAIN



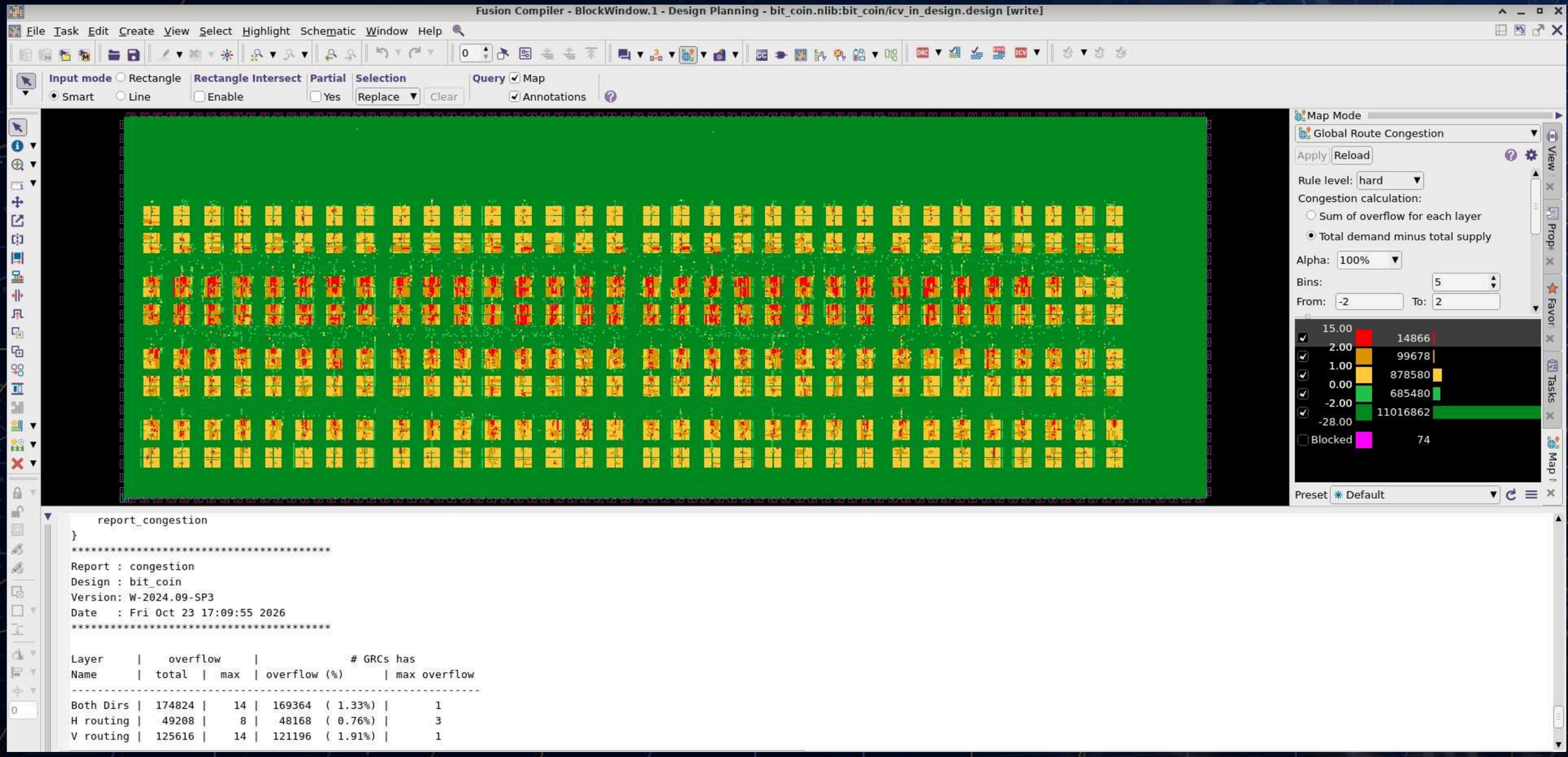
TOP

PD_PISO_SECURE

PD_SIPO_SECURE

DAY 02 • ANALYSIS

CONGESTION REPORT



DAY 03
FREQUENCY



DAY 03 • FREQUENCY

PLAN THE FREQUENCY TESTCASE

Explore feasible performance before deciding on optimisation target

test Case	Frequency	TIME (Required clock period)
Baseline	`1GHz	1.000
testCase 01	1.45GHz	0.689
testCase 02	1.55GHz	0.645
testCase 03	1.63GHZ	0.613
testCase 04	1.80 GHz	0.555
testCase 05	1.90 GHz	0.526
testCase 06	1.95 GHz	0.510
testCase 07	2.0 Ghz	0.500

SWEEP STRATEGY

- **Change one parameter**
  Start with clock period in SDC.
- **Use isolated folders**
  Keep output and logs separate per testcase.
- **Observe feasibility**
  WNS / TNS / NVP + run errors.

DAY 03 • FREQUENCY

FREQUENCY TESTCASE RESULT

Explore feasible performance before deciding on optimisation target

test Case	Frequency	TIME (Required clock period)	WNS (ns)	TNS (ns)	NVP	Cell area	Congestion	Hold / DRV	Verdict
Baseline	1GHz	1.000	0.0005	0.0000	0	base (9.72%)	OK (0.10%)	Hold: 0 violations; DRV: 8,903 nets	Baseline
testCase 01	1.45GHz	0.689	-0.0260	-0.1940	36	9.74%	OK (0.10%)	Hold:1013 DRV:9052	Promising Backup
testCase 02	1.55GHz	0.645	-0.0130	-0.0360	7	9.66%	OK (0.09%)	Hold:0 DRV:8987	BEST
testCase 03	1.63GHZ	0.613	-0.0370	-0.3410	49	9.6	OK (0.07%)	hold:1855 drv:9083	Promising Backup
testCase 04	1.80 GHz	0.555	-0.0620	-3.9960	336	9.82%	OK (0.101%)	DRV: 9032: HOLD: 409	Too high
testCase 05	1.90 GHz	0.526	-0.1920	-67.2000	1,598	9.88%	OK (0.10%)	Hold: 159 violations; DRV: 8,890 nets	Too high
testCase 06	1.95 GHz	0.510	-0.0970	-12.2000	722	9.78%	OK (0.10%)	Hold:863 violations; DRV: 8,825 nets	Too high
testCase 07	2.0 Ghz	0.500	-0.0819	-9.7886	564	9.83%	OK (0.097%)	Hold: 170 violations; DRV: 8,887 nets	Too high

DAY 03 • EXECUTION

DECIDE & TRY PUSH BEST 3 TESTCASES

Controlled experiments enable apples-to-apples QoR comparisons

BASELINE

1.00 GHZ

Original 1G

TEST CASE 01

1.45 GHZ

Isolated folder

TEST CASE 02

1.55 GHZ

Isolated folder

TEST CASE 03

1.63 GHZ

Isolated folder

RUN RULES

- **One directory per clock period**
  Change SDC and rm_user_plugin_scripts, run the entire dependency chain, collect stage logs and compare metrics.

DAY 04
RECIPE CHANGE
(PRE / POST SCRIPTS)

DAY 03 - 04 • EXPERIMENTATION

PRE / POST SCRIPTS (REVISED FREQUENCIES)

Every design change should have a hypothesis and measured effect

Frequency	TIME (Required clock period)	Stage	Controlled Experiment
1.45GHz	0.689	clock_opt_cts_pre_script.tcl	cts.common.max_fanout: 24 -> 16
		route_opt_pre_scripts.tcl	targeted CCD WNS: false -> true
1.45GHz + uncertainty + High Effort	0.689	clock_opt_cts_pre_script.tcl	cts.common.max_fanout: 24 -> 16
		route_opt_pre_scripts.tcl	targeted CCD WNS: false -> true
1.55GHz	0.645	clock_opt_cts_pre_script.tcl	cts.common.max_fanout: 24 -> 16
		route_opt_pre_scripts.tcl	targeted CCD WNS: false -> true
1.55GHz + uncertainty + High Effort	0.645	clock_opt_cts_pre_script.tcl	cts.common.max_fanout: 24 -> 16
		route_opt_pre_scripts.tcl	targeted CCD WNS: false -> true
1.63GHZ	0.613	clock_opt_cts_pre_script.tcl	cts.common.max_fanout: 24 -> 16
		route_opt_pre_scripts.tcl	targeted CCD WNS: false -> true
1.63GHz + uncertainty + High Effort	0.613	clock_opt_cts_pre_script.tcl	cts.common.max_fanout: 24 -> 16
		route_opt_pre_scripts.tcl	targeted CCD WNS: false -> true

DAY 04 • FREQUENCY PUSH TO GET BETTER RESULT

PUSH FREQUENCY RESULTS

Freq	TIME (Required clock period)	WNS (ns)	TNS (ns)	NVP	Cell area	Congestion	Hold / DRV	Skew	Verdict
1.45GHz	0.689	-0.0260	-0.1940	36	9.74% (base)	0.10%	Hold:1013 DRV:9052	0.678 0.609	NORMAL
1.45GHz + High Effort	0.689	-0.0910	-3.6200	205	9.91%	0.10%	Hold:1150 DRV:9180	0.890 0.820	WORSE
1.55GHz	0.645	-0.0140	-0.0040	5	9.66%	0.09%	Hold:0 DRV:9000	0.740 0.722	NORMAL
1.55GHz + High Effort	0.645	-0.0370	-2.9400	165	9.83%	0.11%	Hold:71 DRV:9140	0.950 0.890	WORSE
1.63GHZ	0.613	-0.0370	-0.3410	49	9.60%	0.07%	hold:1855 drv:9083	1.011 0.927	NORMAL
1.63GHz + High Effort	0.613	-0.1420	-14.6000	610	9.88%%	0.12%	Hold:2120 DRV:9340	1.210 1.140	WORSE

Conclusion : After Push with high effort, timing getting worse.

DAY 04 • FREQUENCY PUSH TO GET BETTER RESULT

WHY DID THE TIMING PUSH GET WORSE?

Tighter clock periods + aggressive optimization can worsen setup, hold and electrical QoR.

THE CORE PROBLEM

More effort ≠ better QoR

As the period shrinks, setup margin disappears. Different optimization goals can then fight each other.

CLOCK	Less setup budget
OPTIMIZE	More buffers / routes
RESULT	Hold + DRV can rise

01 CCD / USEFUL SKEW

A reported register self-loop receives little useful-skew benefit; skew changes elsewhere may increase hold risk.

02 HIGH HOLD EFFORT

Extra hold buffers can increase cell area, load and capacitance, potentially raising DRV counts.

03 POWER RECOVERY

Downsizing or HVT swaps reduce power, but may consume already-small setup margin.

04 CROSSTALK REDUCTION

Spacing and routing changes may increase wirelength or delay, changing post-route timing.

DAY 05 • APPLY ECO

DAY 05
APPLY MANUAL ECO

DAY 05 • FREQUENCY PUSH TO GET BETTER RESULT

FINAL DESICION

Freq	TIME (Required clock period)	WNS (ns)	TNS (ns)	NVP	Cell area	Congestion	Hold / DRV	Skew	Verdict
1.45GHz	0.689	-0.0260	-0.1940	36	9.74% (base)	0.10%	Hold:1013 DRV:9052	0.678 0.609	NORMAL
1.45GHz + High Effort	0.689	-0.0910	-3.6200	205	9.91%	0.10%	Hold:1150 DRV:9180	0.890 0.820	WORSE
1.55GHz	0.645	-0.0140	-0.0190	5	9.66%	0.09%	Hold:0 DRV:9000	0.740 0.722	NORMAL
1.55GHz + High Effort	0.645	-0.0370	-2.9400	165	9.83%	0.11%	Hold:71 DRV:9140	0.950 0.890	WORSE
1.63GHZ	0.613	-0.0370	-0.3410	49	9.60%	0.07%	hold:1855 drv:9083	1.011 0.927	NORMAL
1.63GHz + High Effort	0.613	-0.1420	-14.6000	610	9.88%%	0.12%	Hold:2120 DRV:9340	1.210 1.140	WORSE

Conclusion : After Push with high effort, timing getting worse.

DAY 05 • APPLY ECO

QOR OF THE REMAINING VIOLATIONS

Target the remaining failing paths and design-rule checks

SETUP WNS

-0.014 ns

SETUP TNS

-0.019 ns

FAILING ENDPOINT (NVE)

5

NET WITH DRC VIOLATIONS

8,871

pduser11@psdc-eda03:~/labs/bitcoin_pnr_lab_... × pduser11@psdc-eda03:~/labs/bitcoin_pnr_lab_... ×

```
Date   : Sat Oct 31 16:14:46 2026
****************************************
Information: Timer using 'SI, Timing Window Analysis, AWP, POCV-Slew_Variation, CRPR'.
 (TIM-050)

Timing
------------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
------------------------------------------------------------------------------
FUNC_0.78V_SETUP   (Setup)           -0.014          -0.019               5
FUNC_0.85V_SETUP   (Setup)           -0.014          -0.019               5
Design             (Setup)           -0.014          -0.019               5

FUNC_0.85V_HOLD    (Hold)             0.000           0.000               0
Design             (Hold)             0.000           0.000               0
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                          3777808.475
Cell Area (netlist and physical only):        16003345.109
Nets with DRC Violations:     8871
1
fc_shell>
```

1.55GHZ qor report at chip_finish

DAY 05 • APPLY ECO

APPLYING MANUAL ECO METHODS

RVT TO LVT CELL SWAPPING

Improve setup timing by replacing selected Regular Vt (RVT) standard cells with faster Low Vt (LVT) cells using manual ECO in Fusion Compiler.

Path Group :

****reg2out_default****

Change :

RVT → LVT

```
Startpoint: bit_secure_8/sipo_slice_first/wr_reg (rising edge-triggered flip-flop clocked by lclk)
Endpoint: memory_address[33] (output port clocked by lclk)
Mode: FUNC_078
Corner: FUNC_0.78V_SETUP
Scenario: FUNC_0.78V_SETUP
Path Group: **reg2out_default**
Path Type: max
Sigma:   3.00

  Point                                                                                                  Incr      Path
  ---------------------------------------------------------------------------------------------------------------------
  clock lclk (rise edge)                                                                                 0.00      0.00
  clock network delay (propagated)                                                                       1.34      1.34

  bit_secure_8/sipo_slice_first/wr_reg/CLK (RSDFFARX1_LVT)                                               0.00      1.34 r
  bit_secure_8/sipo_slice_first/wr_reg/Q (RSDFFARX1_LVT)                                                 0.17      1.51 f
  bit_secure_8/snps_PD_SIPO_SECURE__ISO_SIPO_OUT_snps_wrISO_SIPO_OUT/Y (LSUPENX1_LVT)                    0.20      1.71 f ~
  bit_secure_8/compile_HFSINV_393_166049/Y (INVX1_LVT)                                                   0.06      1.77 r
  bit_secure_8/ctmi_99/Y (AO22X1_LVT)                                                                    0.07      1.84 r
  compile_HFSINV_2425_151118/Y (IBUFFX8_RVT)                                                             0.09      1.93 f
  compile_HFSINV_2384_151117/Y (IBUFFX8_RVT)                                                             0.13      2.06 r
  compile_HFSINV_2251_151116/Y (IBUFFX8_RVT)                                                             0.12      2.17 f
! compile_HFSINV_2107_151115/Y (INVX8_RVT)                                                               0.09      2.26 r
! compile_HFSINV_1567_151114/Y (INVX8_RVT)                                                               0.08      2.34 f
! compile_HFSINV_1110_151113/Y (INVX8_RVT)                                                               0.09      2.43 r
! compile_HFSINV_698_151112/Y (INVX8_RVT)                                                                0.09      2.53 f
! compile_HFSINV_241_151111/Y (INVX4_LVT)                                                                0.09      2.62 r
! route_opt_ropt_mt_inst_312874/Y (NBUFFX8_RVT)                                                          0.06      2.67 r
! route_opt_ropt_mt_inst_312873/Y (NBUFFX2_LVT)                                                          0.06      2.73 r
! route_opt_ropt_mt_inst_312872/Y (NBUFFX4_RVT)                                                          0.08      2.81 r
! route_opt_ropt_mt_inst_312871/Y (NBUFFX2_RVT)                                                          0.09      2.90 r
! memory_address[33] (out)                                                                              0.00      2.90 r
! data arrival time                                                                                               2.90

  clock lclk (rise edge)                                                                                 1.29      1.29
  clock network delay (propagated)                                                                       1.70      2.99
  clock reconvergence pessimism                                                                          0.00      2.99
  output external delay                                                                                 -0.10      2.89
  data required time                                                                                               2.89
  ---------------------------------------------------------------------------------------------------------------------
  data required time                                                                                               2.89
! data arrival time                                                                                               -2.90
  ---------------------------------------------------------------------------------------------------------------------
! statistical adjustment                                                                                 0.00     -0.01
! slack (VIOLATED)                                                                                               -0.01
```

BEFORE MANUAL ECO

```
Startpoint: bit_secure_8/sipo_slice_first/wr_reg (rising edge-triggered flip-flop clocked by lclk)
Endpoint: memory_address[33] (output port clocked by lclk)
Mode: FUNC_078
Corner: FUNC_0.78V_SETUP
Scenario: FUNC_0.78V_SETUP
Path Group: **reg2out_default**
Path Type: max
Sigma:   3.00

  Point                                                                                                  Incr      Path
  ---------------------------------------------------------------------------------------------------------------------
  clock lclk (rise edge)                                                                                 0.00      0.00
  clock network delay (propagated)                                                                       1.34      1.34

  bit_secure_8/sipo_slice_first/wr_reg/CLK (RSDFFARX1_LVT)                                               0.00      1.34 r
  bit_secure_8/sipo_slice_first/wr_reg/Q (RSDFFARX1_LVT)                                                 0.17      1.51 f
  bit_secure_8/snps_PD_SIPO_SECURE__ISO_SIPO_OUT_snps_wrISO_SIPO_OUT/Y (LSUPENX1_LVT)                    0.20      1.71 f ~
  bit_secure_8/compile_HFSINV_393_166049/Y (INVX1_LVT)                                                   0.06      1.77 r
  bit_secure_8/ctmi_99/Y (AO22X1_LVT)                                                                    0.07      1.84 r
  compile_HFSINV_2425_151118/Y (IBUFFX8_RVT)                                                             0.09      1.93 f
  compile_HFSINV_2384_151117/Y (IBUFFX8_RVT)                                                             0.13      2.06 r
  compile_HFSINV_2251_151116/Y (IBUFFX8_RVT)                                                             0.12      2.17 f
! compile_HFSINV_2107_151115/Y (INVX8_LVT)                                                               0.07      2.25 r
! compile_HFSINV_1567_151114/Y (INVX8_LVT)                                                               0.06      2.30 f
! compile_HFSINV_1110_151113/Y (INVX8_RVT)                                                               0.09      2.39 r
! compile_HFSINV_698_151112/Y (INVX8_LVT)                                                                0.07      2.46 f
! compile_HFSINV_241_151111/Y (INVX4_LVT)                                                                0.09      2.55 r
! route_opt_ropt_mt_inst_312874/Y (NBUFFX8_LVT)                                                          0.05      2.59 r
! route_opt_ropt_mt_inst_312873/Y (NBUFFX2_LVT)                                                          0.06      2.65 r
! route_opt_ropt_mt_inst_312872/Y (NBUFFX4_RVT)                                                          0.08      2.73 r
! route_opt_ropt_mt_inst_312871/Y (NBUFFX2_RVT)                                                          0.09      2.81 r
! memory_address[33] (out)                                                                              0.00      2.82 r
! data arrival time                                                                                               2.82

  clock lclk (rise edge)                                                                                 1.29      1.29
  clock network delay (propagated)                                                                       1.70      2.99
  clock reconvergence pessimism                                                                          0.00      2.99
  output external delay                                                                                 -0.10      2.89
  data required time                                                                                               2.89
  ---------------------------------------------------------------------------------------------------------------------
  data required time                                                                                               2.89
! data arrival time                                                                                               -2.82
  ---------------------------------------------------------------------------------------------------------------------
! statistical adjustment                                                                                 0.00      0.07
! slack (MET)                                                                                                     0.07
```

AFTER MANUAL ECO

DAY 05 • APPLY ECO

Path Group : lclk

Change : RVT → LVT

BEFORE MANUAL ECO

```
Startpoint: bit_secure_15/slice_6/sipo_bit/rd_addr_reg_2_ (rising edge-triggered flip-flop clocked by lclk)
Endpoint: bit_secure_15/slice_6/sipo_bit/rd_addr_reg_3_ (rising edge-triggered flip-flop clocked by lclk)
Mode: FUNC_078
Corner: FUNC_0.78V_SETUP
Scenario: FUNC_0.78V_SETUP
Path Group: lclk
Path Type: max
Sigma:   3.00

Point                                                                                                  Incr      Path
-------------------------------------------------------------------------------------------------------------------------
clock lclk (rise edge)                                                                                 0.00      0.00
clock network delay (propagated)                                                                       1.42      1.42

bit_secure_15/slice_6/sipo_bit/rd_addr_reg_2_/CLK (RSDFFARX1_RVT)                                      0.00      1.42 r
bit_secure_15/slice_6/sipo_bit/rd_addr_reg_2_/QN (RSDFFARX1_RVT)                                       0.24      1.66 f
bit_secure_15/slice_6/sipo_bit/ctmi_341/Y (AND4X1_RVT)                                                 0.13      1.78 f
bit_secure_15/slice_6/sipo_bit/ctmi_351/Y (XNOR2X1_RVT)                                                0.13      1.91 r
bit_secure_15/slice_6/sipo_bit/rd_addr_reg_3_/D (RSDFFARX1_RVT)                                        0.00      1.91 r
data arrival time                                                                                                1.91

clock lclk (rise edge)                                                                                 0.64      0.64
clock network delay (propagated)                                                                       1.40      2.05
clock reconvergence pessimism                                                                          0.00      2.05
bit_secure_15/slice_6/sipo_bit/rd_addr_reg_3_/CLK (RSDFFARX1_RVT)                                      0.00      2.05 r
library setup time                                                                                    -0.13      1.91
data required time                                                                                               1.91
-------------------------------------------------------------------------------------------------------------------------
data required time                                                                                               1.91
data arrival time                                                                                               -1.91
-------------------------------------------------------------------------------------------------------------------------
statistical adjustment                                                                                 0.00     -0.00
slack (VIOLATED)                                                                                                -0.00
```

AFTER MANUAL ECO

```
Startpoint: bit_secure_15/slice_6/sipo_bit/rd_addr_reg_2_ (rising edge-triggered flip-flop clocked by lclk)
Endpoint: bit_secure_15/slice_6/sipo_bit/rd_addr_reg_3_ (rising edge-triggered flip-flop clocked by lclk)
Mode: FUNC_078
Corner: FUNC_0.78V_SETUP
Scenario: FUNC_0.78V_SETUP
Path Group: lclk
Path Type: max
Sigma:   3.00

Point                                                                                                  Incr      Path
-------------------------------------------------------------------------------------------------------------------------
clock lclk (rise edge)                                                                                 0.00      0.00
clock network delay (propagated)                                                                       1.42      1.42

bit_secure_15/slice_6/sipo_bit/rd_addr_reg_2_/CLK (RSDFFARX1_RVT)                                      0.00      1.42 r
bit_secure_15/slice_6/sipo_bit/rd_addr_reg_2_/QN (RSDFFARX1_RVT)                                       0.24      1.66 f
bit_secure_15/slice_6/sipo_bit/ctmi_341/Y (AND4X1_LVT)                                                 0.09      1.74 f
bit_secure_15/slice_6/sipo_bit/ctmi_351/Y (XNOR2X1_LVT)                                                0.08      1.82 r
bit_secure_15/slice_6/sipo_bit/rd_addr_reg_3_/D (RSDFFARX1_RVT)                                        0.00      1.82 r
data arrival time                                                                                                1.82

clock lclk (rise edge)                                                                                 0.64      0.64
clock network delay (propagated)                                                                       1.40      2.05
clock reconvergence pessimism                                                                          0.00      2.05
bit_secure_15/slice_6/sipo_bit/rd_addr_reg_3_/CLK (RSDFFARX1_RVT)                                      0.00      2.05 r
library setup time                                                                                    -0.13      1.92
data required time                                                                                               1.92
-------------------------------------------------------------------------------------------------------------------------
data required time                                                                                               1.92
data arrival time                                                                                               -1.82
-------------------------------------------------------------------------------------------------------------------------
statistical adjustment                                                                                 0.00      0.10
slack (MET)                                                                                                      0.10
```

Path Group : lclk

Change : RVT → LVT

```
Startpoint: bit_secure_3/slice_15/sipo_bit/rd_addr_reg_1_ (rising edge-triggered flip-flop clocked by lclk)
Endpoint: bit_secure_3/slice_15/sipo_bit/rd_addr_reg_3_ (rising edge-triggered flip-flop clocked by lclk)
Mode: FUNC_078
Corner: FUNC_0.78V_SETUP
Scenario: FUNC_0.78V_SETUP
Path Group: lclk
Path Type: max
Sigma:   3.00

Point                                                                                                  Incr      Path
-------------------------------------------------------------------------------------------------------------------------
clock lclk (rise edge)                                                                                 0.00      0.00
clock network delay (propagated)                                                                       1.47      1.47

bit_secure_3/slice_15/sipo_bit/rd_addr_reg_1_/CLK (RSDFFARX1_RVT)                                      0.00      1.47 r
bit_secure_3/slice_15/sipo_bit/rd_addr_reg_1_/QN (RSDFFARX1_RVT)                                       0.25      1.72 f
bit_secure_3/slice_15/sipo_bit/ctmi_341/Y (AND4X1_RVT)                                                 0.13      1.84 f
bit_secure_3/slice_15/sipo_bit/ctmi_351/Y (XNOR2X1_RVT)                                                0.13      1.98 r
bit_secure_3/slice_15/sipo_bit/rd_addr_reg_3_/D (RSDFFARX1_RVT)                                        0.00      1.98 r
data arrival time                                                                                                1.98

clock lclk (rise edge)                                                                                 0.64      0.64
clock network delay (propagated)                                                                       1.47      2.11
clock reconvergence pessimism                                                                          0.00      2.11
bit_secure_3/slice_15/sipo_bit/rd_addr_reg_3_/CLK (RSDFFARX1_RVT)                                      0.00      2.11 r
library setup time                                                                                    -0.14      1.97
data required time                                                                                               1.97
-------------------------------------------------------------------------------------------------------------------------
data required time                                                                                               1.97
data arrival time                                                                                               -1.98
-------------------------------------------------------------------------------------------------------------------------
statistical adjustment                                                                                 0.00     -0.00
slack (VIOLATED)                                                                                                -0.00
```

```
Startpoint: bit_secure_3/slice_15/sipo_bit/rd_addr_reg_0_ (rising edge-triggered flip-flop clocked by lclk)
Endpoint: bit_secure_3/slice_15/sipo_bit/rd_addr_reg_3_ (rising edge-triggered flip-flop clocked by lclk)
Mode: FUNC_078
Corner: FUNC_0.78V_SETUP
Scenario: FUNC_0.78V_SETUP
Path Group: lclk
Path Type: max
Sigma:   3.00

Point                                                                                                  Incr      Path
-------------------------------------------------------------------------------------------------------------------------
clock lclk (rise edge)                                                                                 0.00      0.00
clock network delay (propagated)                                                                       1.47      1.47

bit_secure_3/slice_15/sipo_bit/rd_addr_reg_0_/CLK (RSDFFARX1_RVT)                                      0.00      1.47 r
bit_secure_3/slice_15/sipo_bit/rd_addr_reg_0_/QN (RSDFFARX1_RVT)                                       0.25      1.72 f
bit_secure_3/slice_15/sipo_bit/ctmi_341/Y (AND4X1_LVT)                                                 0.09      1.81 f
bit_secure_3/slice_15/sipo_bit/ctmi_351/Y (XNOR2X1_LVT)                                                0.08      1.89 r
bit_secure_3/slice_15/sipo_bit/rd_addr_reg_3_/D (RSDFFARX1_RVT)                                        0.00      1.89 r
data arrival time                                                                                                1.89

clock lclk (rise edge)                                                                                 0.64      0.64
clock network delay (propagated)                                                                       1.47      2.11
clock reconvergence pessimism                                                                          0.00      2.11
bit_secure_3/slice_15/sipo_bit/rd_addr_reg_3_/CLK (RSDFFARX1_RVT)                                      0.00      2.11 r
library setup time                                                                                    -0.14      1.98
data required time                                                                                               1.98
-------------------------------------------------------------------------------------------------------------------------
data required time                                                                                               1.98
data arrival time                                                                                               -1.89
-------------------------------------------------------------------------------------------------------------------------
statistical adjustment                                                                                 0.00      0.09
slack (MET)                                                                                                      0.09
```

DAY 05 • APPLY ECO

DEFAULT, OPTIMIZED AND FINAL QOR

default.rpt — optimize_qor.rpt — final_qor.rpt

Meld File Edit Changes View

Save Undo

default.rpt ...nal_qor.rpt

default.rpt

```
Max Cap Violations:               8837
----------------------------------------

1
****************************************
Report : qor
        -summary
Design : bit_coin
Version: W-2024.09-SP3
Date   : Sun Nov  1 02:47:11 2026
****************************************
Information: Timer using 'SI, Timing Window Analysis, AWP, POCV-Slew_Variation,

Timing
--------------------------------------------------------------------------------
Context                                WNS          TNS          NVE
--------------------------------------------------------------------------------
FUNC_0.78V_SETUP    (Setup)         -0.013       -0.036            7
FUNC_0.85V_SETUP    (Setup)         -0.013       -0.036            7
Design              (Setup)         -0.013       -0.036            7

FUNC_0.85V_HOLD     (Hold)           0.000        0.000            0
Design              (Hold)           0.000        0.000            0
--------------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------------
Cell Area (netlist):                        3778103.791
Cell Area (netlist and physical only):      16003357.816
Nets with DRC Violations:     8858
1
```

optimize_qor.rpt

```
Max Cap Violations:               8853
----------------------------------------

1
****************************************
Report : qor
        -summary
Design : bit_coin
Version: W-2024.09-SP3
Date   : Fri Oct 30 02:16:07 2026
****************************************
Information: Timer using 'SI, Timing Window Analysis, AWP, POCV-Slew_Variati

Timing
--------------------------------------------------------------------------------
Context                                WNS          TNS          NVE
--------------------------------------------------------------------------------
FUNC_0.78V_SETUP    (Setup)         -0.014       -0.019            5
FUNC_0.85V_SETUP    (Setup)         -0.014       -0.019            5
Design              (Setup)         -0.014       -0.019            5

FUNC_0.85V_HOLD     (Hold)           0.000        0.000            0
Design              (Hold)           0.000        0.000            0
--------------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------------
Cell Area (netlist):                        3777808.475
Cell Area (netlist and physical only):      16003345.109
Nets with DRC Violations:     8871
1
```

final_qor.rpt

```
Max Cap Violations:               8855
----------------------------------------

1
****************************************
Report : qor
        -summary
Design : bit_coin
Version: W-2024.09-SP3
Date   : Fri Oct 30 15:33:14 2026
****************************************
Information: Timer using 'SI, Timing Window Analysis, AWP, POCV-Slew_Variation,

Timing
--------------------------------------------------------------------------------
Context                                WNS          TNS          NVE
--------------------------------------------------------------------------------
FUNC_0.78V_SETUP    (Setup)          0.000        0.000            0
FUNC_0.85V_SETUP    (Setup)          0.000        0.000            0
Design              (Setup)          0.000        0.000            0

FUNC_0.85V_HOLD     (Hold)           0.000        0.000            0
Design              (Hold)           0.000        0.000            0
--------------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------------
Cell Area (netlist):                        3777808.475
Cell Area (netlist and physical only):      16003345.109
Nets with DRC Violations:     8873
1
```

Unicode (UTF-8) Ln 1, Col 1 | Unicode (UTF-8) Ln 5, Col 1 | Unicode (UTF-8) Ln 242, Col 2

1.55 GHZ (DEFAULT) | **1.55 GHZ (OPTIMIZED)** | **1.55 GHZ FINAL**

DAY 05 • APPLY ECO

FREQUENCY SELECTION & ECO OPTIMIZATION

Eight clock targets → Three shortlisted frequencies → Before / after ECO → Final selection

01 INITIAL FREQUENCY EXPLORATION

08 / CLOCK TARGETS

1.00 GHz	1.45 GHz	1.55 GHz	1.63 GHz	1.80 GHz	1.90 GHz	1.95 GHz	2.00 GHz

02 SHORTLISTED CANDIDATES

03 / FREQUENCIES

- **1.45 GHz** — CLOCK PERIOD 0.690 ns
- **1.55 GHz** — CLOCK PERIOD 0.645 ns — ECO
- **1.63 GHz** — CLOCK PERIOD 0.613 ns

03 AFTER OPTIMIZED

1.55 GHz TARGET

1.55 GHz | AFTER OPTIMIZED: Violations present | MOSTLY VIOLATED

RVT → LVT CELL SWAPPING

04 AFTER MANUAL ECO

1.55 GHz TARGET

1.55 GHz | AFTER MANUAL ECO: Target met | MET

05 FINAL FREQUENCY SELECTION

01 / FINAL TARGET

1.55 GHz | FINAL SELECTED FREQUENCY: Clock period ≈ 0.645 ns

DAY 06
VERIFY SCRIPT AND FC VS PT CORRELATION

DAY 06 • CORRELATION

EDA SCRIPTING & REPORT EXTRACTION

Improve repeatability and accelerate analysis across multiple runs

TIMING PARSING

Capture WNS, TNS, violating paths

NETLIST DIFF

Compare design changes between runs

SLOPE & CAP

Report transitions and capacitance

PIN MARGIN

Estimate pin-level timing margin



DAY 06 • CORRELATION

TIMING RESULT

Start point	End point	Slack	Cell Dela	RC Dela	Logic Stag	Logic Ex Buf
bit_secure_3/slice_15/sipo_bit/rd_addr_reg_1_/CLK	bit_secure_3/slice_15/sipo_bit/rd_addr_reg_3_/D	-0.0041	0.5109	0.0001	2	2
bit_secure_15/slice_1/sipo_bit/rd_addr_reg_2_/CLK	bit_secure_15/slice_1/sipo_bit/rd_addr_reg_3_/D	-0.0018	0.4946	0.001	2	2
bit_secure_15/slice_6/sipo_bit/rd_addr_reg_2_/CLK	bit_secure_15/slice_6/sipo_bit/rd_addr_reg_3_/D	-0.0011	0.497	0.0009	2	2
bit_secure_10/slice_3/sipo_bit/rd_addr_reg_2_/CLK	bit_secure_10/slice_3/sipo_bit/rd_addr_reg_3_/D	-0.0004	0.4869	0	2	2
bit_secure_11/slice_27/sipo_bit/rd_addr_reg_1_/CLK	bit_secure_11/slice_27/sipo_bit/rd_addr_reg_3_/D	-0.0003	0.5046	0.0001	2	2
bit_secure_6/slice_22/sipo_bit/rd_addr_reg_1_/CLK	bit_secure_6/slice_22/sipo_bit/rd_addr_reg_3_/D	-0.0001	0.5057	0.0021	2	2
bit_secure_15/slice_9/sync_datafrommem_dut_sync_sync_out_reg_15_/CLK	bit_secure_15/slice_9/piso_bit/temp_reg_15_/D	-0.0001	0.3017	0.0027	2	2
bit_secure_10/slice_25/sipo_bit/rd_addr_reg_2_/CLK	bit_secure_10/slice_25/sipo_bit/rd_addr_reg_3_/D	0.0001	0.4862	0.0002	2	2
bit_secure_0/slice_26/sync_datafrommem_dut_sync_sync_out_reg_11_/CLK	bit_secure_0/slice_26/piso_bit/temp_reg_11_/D	0.0001	0.3409	0.0062	2	2
bit_secure_10/slice_10/sipo_bit/rd_addr_reg_1_/CLK	bit_secure_10/slice_10/sipo_bit/rd_addr_reg_3_/D	0.0002	0.501	0.0001	2	2
bit_secure_3/slice_17/sipo_bit/rd_addr_reg_0_/CLK	bit_secure_3/slice_17/sipo_bit/rd_addr_reg_3_/D	0.0002	0.4848	0	2	2
bit_secure_2/sipo_slice_last/wr_reg/CLK	bit_secure_2/temp_1_reg/D	0.0004	0.4644	0.0137	2	1
bit_secure_15/slice_5/sync_datafrommem_dut_sync_sync_out_reg_5_/CLK	bit_secure_15/slice_5/piso_bit/temp_reg_5_/D	0.0006	0.3165	0.0255	2	2
bit_secure_11/slice_26/sync_datafrommem_dut_sync_sync_out_reg_1_/CLK	bit_secure_11/slice_26/piso_bit/temp_reg_1_/D	0.0007	0.3324	0.0004	2	2
bit_secure_11/slice_23/lreset_sync_reset_sync_reg/CLK	bit_secure_11/slice_23/sipo_bit/tmp_reg_11_/RSTB	0.0008	0.4373	0.0161	4	3
bit_secure_11/slice_23/lreset_sync_reset_sync_reg/CLK	bit_secure_11/slice_23/sipo_bit/tmp_reg_12_/RSTB	0.0008	0.4373	0.0161	4	3
bit_secure_15/slice_4/sync_datafrommem_dut_sync_sync_out_reg_4_/CLK	bit_secure_15/slice_4/piso_bit/temp_reg_4_/D	0.0008	0.3786	0.0103	2	2
bit_secure_15/slice_2/sipo_bit/rd_addr_reg_2_/CLK	bit_secure_15/slice_2/sipo_bit/rd_addr_reg_3_/D	0.0008	0.5005	0.0039	2	2
bit_secure_3/slice_15/sipo_bit/rd_addr_reg_0_/CLK	bit_secure_3/slice_15/sipo_bit/rd_addr_reg_1_/D	0.0009	0.4747	0.0053	2	2
bit_secure_11/slice_23/lreset_sync_reset_sync_reg/CLK	bit_secure_11/slice_23/sipo_bit/tmp_reg_10_/RSTB	0.0009	0.4373	0.016	4	3
bit_secure_11/slice_23/sipo_bit/rd_addr_reg_0_/CLK	bit_secure_11/slice_23/sipo_bit/rd_addr_reg_3_/D	0.0009	0.5052	0.0016	2	2
bit_secure_11/slice_23/sync_datafrommem_dut_sync_sync_out_reg_14_/CLK	bit_secure_11/slice_23/piso_bit/temp_reg_14_/D	0.0009	0.2959	0.0104	2	2
bit_secure_10/slice_14/sync_datafrommem_dut_sync_sync_out_reg_9_/CLK	bit_secure_10/slice_14/piso_bit/temp_reg_9_/D	0.001	0.3503	0.0069	2	2
bit_secure_11/slice_23/lreset_sync_reset_sync_reg/CLK	bit_secure_11/slice_23/sipo_bit/pout_reg_6_/RSTB	0.0011	0.4373	0.0157	4	3
bit_secure_11/slice_23/lreset_sync_reset_sync_reg/CLK	bit_secure_11/slice_23/sipo_bit/pout_reg_7_/RSTB	0.0011	0.4373	0.0157	4	3
bit_secure_10/slice_14/sync_datafrommem_dut_sync_sync_out_reg_7_/CLK	bit_secure_10/slice_14/piso_bit/temp_reg_7_/D	0.0012	0.346	0.0101	2	2
bit_secure_11/slice_24/sipo_bit/rd_addr_reg_1_/CLK	bit_secure_11/slice_24/sipo_bit/rd_addr_reg_3_/D	0.0012	0.5046	0.0001	2	2
bit_secure_3/slice_20/piso_bit/dout_reg/CLK	bit_secure_3/slice_20/piso_bit/dout_reg/D	0.0012	0.4997	0	2	2
bit_secure_3/slice_19/sipo_bit/rd_addr_reg_1_/CLK	bit_secure_3/slice_19/sipo_bit/rd_addr_reg_3_/D	0.0013	0.4938	0	2	2
bit_secure_3/slice_14/sipo_bit/rd_addr_reg_0_/CLK	bit_secure_3/slice_14/sipo_bit/rd_addr_reg_3_/D	0.0013	0.5061	0.0001	2	2
bit_secure_11/slice_23/lreset_sync_reset_sync_reg/CLK	bit_secure_11/slice_23/sipo_bit/tmp_reg_4_/RSTB	0.0013	0.4373	0.0157	4	3
bit_secure_10/slice_14/sync_datafrommem_dut_sync_sync_out_reg_13_/CLK	bit_secure_10/slice_14/piso_bit/temp_reg_13_/D	0.0016	0.3515	0.0049	2	2
bit_secure_3/slice_16/sipo_bit/rd_addr_reg_2_/CLK	bit_secure_3/slice_16/sipo_bit/rd_addr_reg_3_/D	0.0017	0.4924	0.0016	2	2
bit_secure_11/slice_30/sipo_bit/rd_addr_reg_2_/CLK	bit_secure_11/slice_30/sipo_bit/rd_addr_reg_3_/D	0.0017	0.4943	0.0001	2	2
bit_secure_3/slice_21/sipo_bit/rd_addr_reg_2_/CLK	bit_secure_3/slice_21/sipo_bit/rd_addr_reg_3_/D	0.0018	0.4861	0.002	2	2
bit_secure_10/slice_14/sync_datafrommem_dut_sync_sync_out_reg_11_/CLK	bit_secure_10/slice_14/piso_bit/temp_reg_11_/D	0.0018	0.3518	0.0056	2	2
bit_secure_3/slice_4/sipo_bit/rd_addr_reg_1_/CLK	bit_secure_3/slice_4/sipo_bit/rd_addr_reg_3_/D	0.0019	0.5085	0.0014	2	2
bit_secure_15/slice_4/sync_datafrommem_dut_sync_sync_out_reg_10_/CLK	bit_secure_15/slice_4/piso_bit/temp_reg_10_/D	0.002	0.3719	0.0182	2	2
bit_secure_10/slice_14/sync_datafrommem_dut_sync_sync_out_reg_2_/CLK	bit_secure_10/slice_14/piso_bit/temp_reg_2_/D	0.002	0.3442	0.0131	2	2
bit_secure_11/slice_31/sipo_bit/rd_addr_reg_0_/CLK	bit_secure_11/slice_31/sipo_bit/rd_addr_reg_3_/D	0.0021	0.5042	0.0001	2	2
bit_secure_11/slice_30/sync_datafrommem_dut_sync_sync_out_reg_13_/CLK	bit_secure_11/slice_30/piso_bit/temp_reg_13_/D	0.0021	0.332	0.0069	2	2
bit_secure_6/slice_20/sipo_bit/rd_addr_reg_0_/CLK	bit_secure_6/slice_20/sipo_bit/rd_addr_reg_3_/D	0.0022	0.5152	0.0001	4	4
bit_secure_0/slice_30/sync_datafrommem_dut_sync_sync_out_reg_0_/CLK	bit_secure_0/slice_30/piso_bit/temp_reg_0_/D	0.0024	0.3686	0.0017	3	2
bit_secure_4/slice_13/sync_datafrommem_dut_sync_sync_out_reg_12_/CLK	bit_secure_4/slice_13/piso_bit/temp_reg_12_/D	0.0024	0.5834	0.013	3	3
bit_secure_15/slice_5/sync_datafrommem_dut_sync_sync_out_reg_7_/CLK	bit_secure_15/slice_5/piso_bit/temp_reg_7_/D	0.0024	0.32	0.0211	2	2

fc_shell

DAY 06 • CORRELATION

NETLISTDIFF RESULT

	A	B	C	D	E	F	G	H	I	
1	category	cell	ref_A	ref_B	x_A	y_A	x_B	y_B	distance	detail
2	moved	bit_secure_0/load_data_from_memory_0_reg	DFFARX1_RVT	DFFARX1_RVT	1098.648	1502.968	1094.392	1506.312	5.413	dx=-4.256 dy=3.344
3	orient_change	bit_secure_0/load_data_from_memory_0_reg	DFFARX1_RVT	DFFARX1_RVT	1098.648	1502.968	1094.392	1506.312		MY -> MX
4	conn_change	bit_secure_0/load_data_from_memory_0_reg	DFFARX1_RVT	DFFARX1_RVT	1098.648	1502.968	1094.392	1506.312		CLK:bit_secure_0/ls_13093->bit_secure_0/clock_opt_cts_ZCTSNET_1
5	moved	lreset_sync/reset_sync_reg	DFFARX1_RVT	DFFARX1_RVT	3276.656	687.032	3272.4	687.032	4.256	dx=-4.256 dy=0.000
6	orient_change	lreset_sync/reset_sync_reg	DFFARX1_RVT	DFFARX1_RVT	3276.656	687.032	3272.4	687.032		MY -> R0
7	moved	bit_secure_0/load_data_from_memory_1_reg	DFFARX1_RVT	DFFARX1_RVT	1098.344	1506.312	1098.648	1506.312	0.304	dx=0.304 dy=0.000
8	conn_change	bit_secure_0/load_data_from_memory_1_reg	DFFARX1_RVT	DFFARX1_RVT	1098.344	1506.312	1098.648	1506.312		CLK:bit_secure_0/ls_13093->bit_secure_0/clock_opt_cts_ZCTSNET_1
9	moved	hreset_sync/reset_sync_reg	DFFARX1_RVT	DFFARX1_RVT	3533.08	576.68	3528.824	576.68	4.256	dx=-4.256 dy=0.000
10	orient_change	hreset_sync/reset_sync_reg	DFFARX1_RVT	DFFARX1_RVT	3533.08	576.68	3528.824	576.68		MY -> R0
11	moved	bit_secure_10/snps_PD_PISO_SECURE__ISO_PISO_OUT_snps_dout_0ISO_PISO_OUT	LSUPENX1_RVT	LSUPENX1_RVT	2221.928	1504.64	2223.6	1504.64	1.672	dx=1.672 dy=0.000
12	moved	piso_secure_0/temp_reg_15_	RSDFFARX1_RVT	RSDFFARX1_RVT	5020.248	526.52	5009.456	526.52	10.792	dx=-10.792 dy=0.000
13	orient_change	piso_secure_0/temp_reg_15_	RSDFFARX1_RVT	RSDFFARX1_RVT	5020.248	526.52	5009.456	526.52		R180 -> MX
14	conn_change	piso_secure_0/temp_reg_15_	RSDFFARX1_RVT	RSDFFARX1_RVT	5020.248	526.52	5009.456	526.52		CLK:piso_secure_0/clk->piso_secure_0/clock_opt_cts_ZCTSNET_0
15	moved	piso_secure_1/temp_reg_15_	RSDFFARX1_RVT	RSDFFARX1_RVT	6071.328	526.52	6060.536	526.52	10.792	dx=-10.792 dy=0.000
16	orient_change	piso_secure_1/temp_reg_15_	RSDFFARX1_RVT	RSDFFARX1_RVT	6071.328	526.52	6060.536	526.52		R180 -> MX
17	conn_change	piso_secure_1/temp_reg_15_	RSDFFARX1_RVT	RSDFFARX1_RVT	6071.328	526.52	6060.536	526.52		CLK:piso_secure_1/clk->piso_secure_1/clock_opt_cts_ZCTSNET_0
18	moved	bit_secure_15/piso_slice_last/dout_reg	RSDFFARX1_RVT	RSDFFARX1_RVT	5956.112	549.928	5945.32	549.928	10.792	dx=-10.792 dy=0.000
19	orient_change	bit_secure_15/piso_slice_last/dout_reg	RSDFFARX1_RVT	RSDFFARX1_RVT	5956.112	549.928	5945.32	549.928		R180 -> MX
20	moved	bit_secure_15/slice_31/sync_add2mem_dut_sync_sync_out_reg_3_	DFFARX1_RVT	DFFARX1_RVT	6521.552	269.032	6517.296	269.032	4.256	dx=-4.256 dy=0.000
21	orient_change	bit_secure_15/slice_31/sync_add2mem_dut_sync_sync_out_reg_3_	DFFARX1_RVT	DFFARX1_RVT	6521.552	269.032	6517.296	269.032		MY -> R0
22	conn_change	bit_secure_15/slice_31/sync_add2mem_dut_sync_sync_out_reg_3_	DFFARX1_RVT	DFFARX1_RVT	6521.552	269.032	6517.296	269.032		CLK:bit_secure_15/slice_31/hclk->bit_secure_15/slice_31/clock_opt_cts_ZCTSNET_3
23	moved	sync_sout_0/dut_sync_sync_out_reg_1_	DFFARX1_RVT	DFFARX1_RVT	3125.72	576.68	3121.464	576.68	4.256	dx=-4.256 dy=0.000
24	orient_change	sync_sout_0/dut_sync_sync_out_reg_1_	DFFARX1_RVT	DFFARX1_RVT	3125.72	576.68	3121.464	576.68		MY -> R0
25	moved	bit_secure_1/load_data_from_memory_0_reg	DFFARX1_RVT	DFFARX1_RVT	970.512	1459.496	966.256	1459.496	4.256	dx=-4.256 dy=0.000
26	orient_change	bit_secure_1/load_data_from_memory_0_reg	DFFARX1_RVT	DFFARX1_RVT	970.512	1459.496	966.256	1459.496		R180 -> MX
27	conn_change	bit_secure_1/load_data_from_memory_0_reg	DFFARX1_RVT	DFFARX1_RVT	970.512	1459.496	966.256	1459.496		CLK:bit_secure_1/ls_13060->bit_secure_1/clock_opt_cts_ZCTSNET_0
28	moved	sync_sout_0/dut_sync_sync_out_reg_0_	DFFARX1_RVT	DFFARX1_RVT	3731.896	580.024	3727.64	580.024	4.256	dx=-4.256 dy=0.000
29	orient_change	sync_sout_0/dut_sync_sync_out_reg_0_	DFFARX1_RVT	DFFARX1_RVT	3731.896	580.024	3727.64	580.024		MY -> R0
30	conn_change	sync_sout_0/dut_sync_sync_out_reg_0_	DFFARX1_RVT	DFFARX1_RVT	3731.896	580.024	3727.64	580.024		CLK:sync_sout_0/clk->sync_sout_0/clock_opt_cts_ZCTSNET_1
31	moved	sync_sout_0/dut_sync_t_sync_out_reg_0_	SDFFARX1_RVT	SDFFARX1_RVT	3740.864	583.368	3735.392	583.368	5.472	dx=-5.472 dy=0.000
32	orient_change	sync_sout_0/dut_sync_t_sync_out_reg_0_	SDFFARX1_RVT	SDFFARX1_RVT	3740.864	583.368	3735.392	583.368		MY -> R0
33	conn_change	sync_sout_0/dut_sync_t_sync_out_reg_0_	SDFFARX1_RVT	SDFFARX1_RVT	3740.864	583.368	3735.392	583.368		CLK:sync_sout_0/clk->sync_sout_0/clock_opt_cts_ZCTSNET_1
34	moved	sync_sout_0/dut_sync_t_sync_out_reg_1_	SDFFARX1_RVT	SDFFARX1_RVT	3577.008	580.024	3571.536	580.024	5.472	dx=-5.472 dy=0.000
35	orient_change	sync_sout_0/dut_sync_t_sync_out_reg_1_	SDFFARX1_RVT	SDFFARX1_RVT	3577.008	580.024	3571.536	580.024		R180 -> MX
36	conn_change	sync_sout_0/dut_sync_t_sync_out_reg_1_	SDFFARX1_RVT	SDFFARX1_RVT	3577.008	580.024	3571.536	580.024		CLK:sync_sout_0/clk->sync_sout_0/clock_opt_cts_ZCTSNET_0
37	conn_change	bit_secure_15/slice_31/lreset_sync_t_reset_sync_reg	SDFFARX1_RVT	SDFFARX1_RVT	6427.312	566.648	6427.312	566.648		CLK:bit_secure_15/slice_31/ls_12599->bit_secure_15/slice_31/clock_opt_cts_ZCTSNET_0
38	moved	sync_hash_data/dut_sync_t_sync_out_reg_11_	SDFFARX1_RVT	SDFFARX1_RVT	2761.68	680.344	2756.208	680.344	5.472	dx=-5.472 dy=0.000
39	orient_change	sync_hash_data/dut_sync_t_sync_out_reg_11_	SDFFARX1_RVT	SDFFARX1_RVT	2761.68	680.344	2756.208	680.344		R180 -> MX
40	conn_change	sync_hash_data/dut_sync_t_sync_out_reg_11_	SDFFARX1_RVT	SDFFARX1_RVT	2761.68	680.344	2756.208	680.344		D:sync_hash_data/din[11]->sync_hash_data/clock_opt_cts_opt_ZINV_4_10 CLK:sync_hash_data/clk->sync_
41	moved	sync_hash_data/dut_sync_t_sync_out_reg_10_	SDFFARX1_RVT	SDFFARX1_RVT	2749.672	677	2744.2	677	5.472	dx=-5.472 dy=0.000
42	orient_change	sync_hash_data/dut_sync_t_sync_out_reg_10_	SDFFARX1_RVT	SDFFARX1_RVT	2749.672	677	2744.2	677		R180 -> MX
43	conn_change	sync_hash_data/dut_sync_t_sync_out_reg_10_	SDFFARX1_RVT	SDFFARX1_RVT	2749.672	677	2744.2	677		D:sync_hash_data/din[10]->sync_hash_data/clock_opt_cts_opt_ZINV_4_4 CLK:sync_hash_data/clk->sync_h

fc_shell

```
================================================================
       DAY 06 - FC vs PT NETLIST DIFF RESULT
================================================================
Design           : bit_coin
Stage            : icv_in_design
Comparison       : Fusion Compiler vs PrimeTime

----------------------------------------------------------------
1. ORIGINAL NETLIST CELL COUNT
----------------------------------------------------------------
Fusion Compiler :    1,887,693
PrimeTime       :      191,078
Difference      :    1,696,615

----------------------------------------------------------------
2. ROOT CAUSE - FC-ONLY PHYSICAL FILLER CELLS
----------------------------------------------------------------
Filler Reference                              Count
----------------------------------------------------------------
SHFILL3_RVT                               1,206,382
SHFILL128_RVT                               330,075
SHFILL1_RVT                                  70,087
SHFILL2_LVT                                  57,807
SHFILL64_RVT                                 31,970
SHFILL3_LVT                                    181
DHFILLHL2_LVT                                   65
SHFILL128_LVT                                   21
DHFILLHLHLS11_LVT                               10
SHFILL64_LVT                                     8
SHFILL1_LVT                                      8
DHFILLH2_LVT                                     1
----------------------------------------------------------------
TOTAL FC-ONLY FILLERS                     1,696,615
FC-ONLY NON-FILLER CELLS                          0
PT-ONLY CELLS                                     0

----------------------------------------------------------------
3. FILTERED NETLIST COMPARISON
----------------------------------------------------------------
FC without fillers    :      191,078
PT cell count         :      191,078
Matching instances    :      191,078
Reference mismatches  :            0
Missing in PT         :            0
Extra in PT           :            0

================================================================
FINAL RESULT: MATCH - EXCLUDING PHYSICAL FILLER CELLS
================================================================

CONCLUSION:
The original cell-count difference is explained by
FC-only physical filler instances.
After excluding fillers, compare matching cell names
```

pt_shell

DAY 06 • CORRELATION

INPUT/OUTPUT SLOPE AND OUTPUT CAP RESULT

Mode: CELL

Clock_domain	Path_type	Pin	Lib_cell	Rise_slope	Fall_slope	Cap
lclk	reg2po	compile_HFSBUF_485_133670/Y	NBUFFX32_LVT	0.0366	0.0360	141.0660
lclk	reg2po	compile_HFSBUF_923_161418/Y	NBUFFX32_LVT	0.0370	0.0363	140.1797
lclk	reg2po	bit_secure_7/piso_slice_last/compile_HFSBUF_66_161614/Y	NBUFFX32_LVT	0.0286	0.0299	139.4334
lclk	reg2po	clock_opt_opto_ZBUF_447_inst_288322/Y	NBUFFX32_RVT	0.0273	0.0253	139.4141
lclk	reg2reg	bit_secure_1/piso_slice_first/compile_HFSBUF_64_161578/Y	NBUFFX32_RVT	0.0292	0.0301	138.7960
hclk	pi2reg	compile_HFSINV_40458_169263/Y	INVX32_RVT	0.0454	0.0461	138.1943
lclk	reg2po	compile_HFSBUF_439_161478/Y	NBUFFX32_RVT	0.0382	0.0329	136.8594
lclk	reg2po	bit_secure_7/piso_slice_first/compile_HFSBUF_48_161612/Y	NBUFFX32_LVT	0.0287	0.0300	136.6106
hclk	pi2reg	compile_ZBUF_1425_inst_237249/Y	NBUFFX32_RVT	0.0354	0.0363	134.8417
lclk	reg2po	compile_ZBUF_2820_inst_211537/Y	NBUFFX32_LVT	0.0327	0.0308	134.5339
lclk	reg2po	clock_opt_opto_ctmTdsLR_1_288144/Y	INVX32_RVT	0.0442	0.0403	134.0278
lclk	reg2po	compile_HFSINV_92_133668/Y	INVX32_RVT	0.2100	0.1885	132.3810
lclk	reg2po	compile_HFSBUF_79_132787/Y	NBUFFX32_RVT	0.0386	0.0385	167.9211
lclk	pi2po	compile_HFSBUF_301_167072/Y	NBUFFX32_RVT	0.0363	0.0364	163.8100
lclk	reg2po	compile_HFSBUF_255_145133/Y	NBUFFX32_RVT	0.0270	0.0230	163.5393
lclk	reg2po	clock_opt_opto_ZBUF_47_inst_288300/Y	NBUFFX32_LVT	0.0617	0.0639	161.2473
lclk	reg2po	compile_HFSBUF_87_161435/Y	NBUFFX32_LVT	0.0352	0.0357	161.0745
lclk	reg2po	clock_opt_opto_ZBUF_532_inst_288302/Y	NBUFFX32_RVT	0.0301	0.0282	128.5375
lclk	reg2po	clock_opt_opto_ZBUF_36_inst_288310/Y	NBUFFX32_LVT	0.0303	0.0286	127.8503
hclk	clk_net	clock_opt_cts_ZCTSINV_861205_278543/Y	INVX32_LVT	0.0596	0.0597	125.7077
lclk	reg2po	compile_ctmTdsLR_1_211363/Y	INVX32_RVT	0.0959	0.0885	125.2967
lclk	reg2po	compile_ZINV_3193_inst_210394/Y	IBUFFX32_RVT	0.0222	0.0207	151.7708
lclk	reg2po	compile_HFSBUF_270_161542/Y	NBUFFX32_RVT	0.0368	0.0380	150.0233
lclk	reg2po	compile_HFSBUF_3218_151080/Y	NBUFFX32_LVT	0.0378	0.0386	148.8974
hclk	pi2reg	compile_HFSBUF_2087_171061/Y	NBUFFX32_LVT	0.0336	0.0324	148.0179
lclk	reg2po	compile_HFSINV_641_134843/Y	INVX32_RVT	0.0603	0.0528	144.0063
lclk	reg2po	compile_ZBUF_9_inst_213725/Y	NBUFFX32_RVT	0.0337	0.0372	143.3873
lclk	reg2po	compile_HFSBUF_74_161547/Y	NBUFFX32_RVT	0.0429	0.0417	141.2032
hclk	pi2reg	compile_HFSBUF_63773_169290/Y	NBUFFX32_LVT	0.1490	0.1546	123.4150
lclk	reg2po	compile_ZBUF_76_inst_211551/Y	NBUFFX32_RVT	0.0651	0.0638	123.1391
hclk	pi2reg	compile_HFSBUF_1280_185760/Y	NBUFFX32_LVT	0.0355	0.0352	122.3470
hclk	pi2reg	compile_HFSBUF_476_185867/Y	NBUFFX32_LVT	0.0446	0.0484	121.6596
hclk	pi2reg	compile_HFSBUF_556_185860/Y	NBUFFX32_LVT	0.0398	0.0416	121.6118
hclk	pi2reg	compile_ZBUF_1106_inst_212651/Y	NBUFFX32_RVT	0.0450	0.0495	132.0133
lclk	reg2po	compile_HFSINV_3705_152752/Y	IBUFFX32_LVT	0.0215	0.0202	131.8566
lclk	reg2po	compile_HFSBUF_37_145135/Y	NBUFFX32_LVT	0.0440	0.0455	131.3211
lclk	reg2po	clock_opt_opto_ZBUF_648_inst_288324/Y	NBUFFX32_RVT	0.0304	0.0294	129.5498
lclk	reg2po	compile_HFSBUF_461_161493/Y	NBUFFX32_LVT	0.0388	0.0340	128.6442
lclk	reg2po	compile_HFSBUF_733_152741/Y	NBUFFX32_RVT	0.0396	0.0399	124.3853
lclk	reg2po	compile_ZINV_346_inst_213001/Y	IBUFFX32_LVT	0.0454	0.0424	123.6093
lclk	reg2po	compile_HFSBUF_1891_152750/Y	NBUFFX32_LVT	0.0835	0.0879	159.6465
lclk	reg2po	compile_HFSBUF_413_134851/Y	NBUFFX32_RVT	0.0343	0.0337	154.9707
lclk	reg2po	compile_HFSBUF_601_161467/Y	NBUFFX32_LVT	0.0356	0.0342	153.6487
lclk	reg2po	clock_opt_opto_ZBUF_140_inst_288298/Y	NBUFFX32_LVT	0.0325	0.0317	152.6994

pt_shell

Mode: CELL

Clock_domain	Path_type	Pin	Lib_cell	Rise_slope	Fall_slope	Cap
lclk	reg2po	compile_HFSBUF_79_132787/Y	NBUFFX32_RVT	0.0388	0.0386	168.3729
lclk	pi2po	compile_HFSBUF_301_167072/Y	NBUFFX32_RVT	0.0363	0.0365	164.3736
lclk	reg2po	compile_HFSBUF_255_145133/Y	NBUFFX32_RVT	0.0270	0.0230	163.9663
lclk	reg2po	compile_HFSBUF_87_161435/Y	NBUFFX32_LVT	0.0353	0.0358	161.4111
lclk	reg2po	clock_opt_opto_ZBUF_47_inst_288300/Y	NBUFFX32_LVT	0.0617	0.0639	161.3532
lclk	reg2po	compile_HFSBUF_1891_152750/Y	NBUFFX32_LVT	0.0834	0.0880	159.6913
lclk	reg2po	compile_HFSBUF_413_134851/Y	NBUFFX32_RVT	0.0342	0.0337	155.1288
lclk	reg2po	compile_HFSBUF_601_161467/Y	NBUFFX32_LVT	0.0355	0.0341	155.0702
lclk	reg2po	clock_opt_opto_ZBUF_140_inst_288298/Y	NBUFFX32_LVT	0.0325	0.0317	152.9138
lclk	reg2po	compile_HFSBUF_1097_161448/Y	NBUFFX32_LVT	0.0264	0.0251	152.6258
lclk	reg2po	compile_ZINV_3193_inst_210394/Y	IBUFFX32_RVT	0.0222	0.0208	152.1759
lclk	reg2po	compile_HFSBUF_270_161542/Y	NBUFFX32_RVT	0.0368	0.0381	150.5637
lclk	reg2po	compile_HFSBUF_3218_151080/Y	NBUFFX32_LVT	0.0378	0.0386	148.9524
hclk	pi2reg	compile_HFSBUF_2087_171061/Y	NBUFFX32_LVT	0.0338	0.0324	148.0683
lclk	reg2po	compile_HFSINV_641_134843/Y	INVX32_RVT	0.0601	0.0528	144.0835
lclk	reg2po	compile_ZBUF_9_inst_213725/Y	NBUFFX32_RVT	0.0337	0.0372	143.5237
lclk	reg2po	compile_HFSBUF_74_161547/Y	NBUFFX32_RVT	0.0429	0.0417	141.7782
lclk	reg2po	compile_HFSBUF_485_133670/Y	NBUFFX32_LVT	0.0366	0.0360	141.2930
lclk	reg2po	compile_HFSBUF_923_161418/Y	NBUFFX32_LVT	0.0370	0.0363	140.2243
lclk	reg2po	clock_opt_opto_ZBUF_447_inst_288322/Y	NBUFFX32_RVT	0.0273	0.0254	139.7764
lclk	reg2po	bit_secure_7/piso_slice_last/compile_HFSBUF_66_161614/Y	NBUFFX32_LVT	0.0287	0.0299	139.5734
lclk	reg2reg	bit_secure_1/piso_slice_first/compile_HFSBUF_64_161578/Y	NBUFFX32_RVT	0.0292	0.0302	139.2886
hclk	pi2reg	compile_HFSINV_40458_169263/Y	INVX32_RVT	0.0456	0.0462	138.4811
lclk	reg2po	compile_HFSBUF_439_161478/Y	NBUFFX32_RVT	0.0381	0.0329	137.4457
lclk	reg2po	bit_secure_7/piso_slice_first/compile_HFSBUF_48_161612/Y	NBUFFX32_LVT	0.0287	0.0300	136.6401
hclk	pi2reg	compile_ZBUF_1425_inst_237249/Y	NBUFFX32_RVT	0.0354	0.0364	135.0198
lclk	reg2po	compile_ZBUF_2820_inst_211537/Y	NBUFFX32_LVT	0.0327	0.0308	134.5095
lclk	reg2po	clock_opt_opto_ctmTdsLR_1_288144/Y	INVX32_RVT	0.0442	0.0402	134.4017
lclk	reg2po	compile_HFSINV_92_133668/Y	INVX32_RVT	0.2104	0.1873	132.5998
hclk	pi2reg	compile_ZBUF_1106_inst_212651/Y	NBUFFX32_RVT	0.0451	0.0496	132.2354
lclk	reg2po	compile_HFSINV_3705_152752/Y	IBUFFX32_LVT	0.0215	0.0202	131.8645
lclk	reg2po	compile_HFSBUF_37_145135/Y	NBUFFX32_LVT	0.0442	0.0457	131.8440
lclk	reg2po	clock_opt_opto_ZBUF_648_inst_288324/Y	NBUFFX32_RVT	0.0304	0.0294	129.5936
lclk	reg2po	compile_HFSBUF_461_161493/Y	NBUFFX32_LVT	0.0389	0.0341	128.9681
lclk	reg2po	clock_opt_opto_ZBUF_532_inst_288302/Y	NBUFFX32_RVT	0.0301	0.0281	128.8593
lclk	reg2po	clock_opt_opto_ZBUF_36_inst_288310/Y	NBUFFX32_LVT	0.0303	0.0286	127.8532
hclk	clk_net	clock_opt_cts_ZCTSINV_861205_278543/Y	INVX32_LVT	0.0596	0.0597	125.7187
lclk	reg2po	compile_ctmTdsLR_1_211363/Y	INVX32_RVT	0.0959	0.0886	125.5109
lclk	reg2po	compile_HFSBUF_733_152741/Y	NBUFFX32_RVT	0.0396	0.0400	124.5104
lclk	reg2po	compile_ZINV_346_inst_213001/Y	IBUFFX32_LVT	0.0455	0.0425	123.7429

Showing 40 of 200902 matching pins (278498 output pins total), sorted by cap (largest first)

fc_shell

DAY 06 • CORRELATION

PIN - MARGIN RESULT

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	PT	bit_coin	CURRENT	compile_ZINV_605_inst_214216	INVX0_RVT	compile_ZINV_605_inst_214216/A	in	SETUP	RISE	max_rise_slack	9.141545	MET	
2	PT	bit_coin	CURRENT	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/A	in	SETUP	FALL	max_fall_slack	9.345277	MET	
3	PT	bit_coin	CURRENT	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/A	in	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
4	PT	bit_coin	CURRENT	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/A	in	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
5	PT	bit_coin	CURRENT	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/A	out	SETUP	RISE	max_rise_slack	9.345276	MET	
6	PT	bit_coin	CURRENT	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/Y	out	SETUP	FALL	max_fall_slack	9.141546	MET	
7	PT	bit_coin	CURRENT	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/Y	out	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
8	PT	bit_coin	CURRENT	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/Y	out	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
9	PT	bit_coin	CURRENT	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/Y	in	SETUP	RISE	max_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
10	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_first/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/A	in	SETUP	FALL	max_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
11	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_first/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/A	in	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
12	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_first/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/A	in	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
13	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_first/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/A	out	SETUP	RISE	max_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
14	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_first/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/Y	out	SETUP	FALL	max_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
15	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_first/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/Y	out	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
16	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_first/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/Y	out	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
17	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_first/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/Y	in	SETUP	RISE	max_rise_slack	0.511034	MET	
18	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_last/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/A	in	SETUP	FALL	max_fall_slack	0.528099	MET	
19	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_last/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/A	in	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
20	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_last/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/A	in	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
21	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_last/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/A	out	SETUP	RISE	max_rise_slack	0.528098	MET	
22	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_last/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/Y	out	SETUP	FALL	max_fall_slack	0.511034	MET	
23	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_last/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/Y	out	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
24	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_last/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/Y	out	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
25	PT	bit_coin	CURRENT	bit_secure_0/sipo_slice_last/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/Y	in	SETUP	RISE	max_rise_slack	0.511034	MET	
26	PT	bit_coin	CURRENT	bit_secure_0/slice_0/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_0/sipo_bit/ctmi_360/A	in	SETUP	FALL	max_fall_slack	0.528099	MET	
27	PT	bit_coin	CURRENT	bit_secure_0/slice_0/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_0/sipo_bit/ctmi_360/A	in	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
28	PT	bit_coin	CURRENT	bit_secure_0/slice_0/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_0/sipo_bit/ctmi_360/A	in	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
29	PT	bit_coin	CURRENT	bit_secure_0/slice_0/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_0/sipo_bit/ctmi_360/A	out	SETUP	RISE	max_rise_slack	0.528099	MET	
30	PT	bit_coin	CURRENT	bit_secure_0/slice_0/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_0/sipo_bit/ctmi_360/Y	out	SETUP	FALL	max_fall_slack	0.511034	MET	
31	PT	bit_coin	CURRENT	bit_secure_0/slice_0/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_0/sipo_bit/ctmi_360/Y	out	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
32	PT	bit_coin	CURRENT	bit_secure_0/slice_0/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_0/sipo_bit/ctmi_360/Y	out	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
33	PT	bit_coin	CURRENT	bit_secure_0/slice_0/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_0/sipo_bit/ctmi_360/Y	in	SETUP	RISE	max_rise_slack	6.275134	MET	
34	PT	bit_coin	CURRENT	bit_secure_0/slice_1/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_1/sipo_bit/ctmi_360/A	in	SETUP	FALL	max_fall_slack	6.318745	MET	
35	PT	bit_coin	CURRENT	bit_secure_0/slice_1/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_1/sipo_bit/ctmi_360/A	in	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
36	PT	bit_coin	CURRENT	bit_secure_0/slice_1/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_1/sipo_bit/ctmi_360/A	in	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
37	PT	bit_coin	CURRENT	bit_secure_0/slice_1/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_1/sipo_bit/ctmi_360/A	out	SETUP	RISE	max_rise_slack	6.318745	MET	
38	PT	bit_coin	CURRENT	bit_secure_0/slice_1/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_1/sipo_bit/ctmi_360/Y	out	SETUP	FALL	max_fall_slack	6.275134	MET	
39	PT	bit_coin	CURRENT	bit_secure_0/slice_1/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_1/sipo_bit/ctmi_360/Y	out	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
40	PT	bit_coin	CURRENT	bit_secure_0/slice_1/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_1/sipo_bit/ctmi_360/Y	out	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
41	PT	bit_coin	CURRENT	bit_secure_0/slice_1/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_1/sipo_bit/ctmi_360/Y	in	SETUP	RISE	max_rise_slack	6.035374	MET	
42	PT	bit_coin	CURRENT	bit_secure_0/slice_10/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_10/sipo_bit/ctmi_360/A	in	SETUP	FALL	max_fall_slack	6.091708	MET	
43	PT	bit_coin	CURRENT	bit_secure_0/slice_10/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_10/sipo_bit/ctmi_360/A	in	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
44	PT	bit_coin	CURRENT	bit_secure_0/slice_10/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_10/sipo_bit/ctmi_360/A	in	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK
45	PT	bit_coin	CURRENT	bit_secure_0/slice_10/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_10/sipo_bit/ctmi_360/A	out	SETUP	RISE	max_rise_slack	6.091708	MET	
46	PT	bit_coin	CURRENT	bit_secure_0/slice_10/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_10/sipo_bit/ctmi_360/Y	out	SETUP	FALL	max_fall_slack	6.035374	MET	
47	PT	bit_coin	CURRENT	bit_secure_0/slice_10/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_10/sipo_bit/ctmi_360/Y	out	HOLD	RISE	min_rise_slack		UNAVAILABLE	NO_FINITE_SLACK
48	PT	bit_coin	CURRENT	bit_secure_0/slice_10/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/slice_10/sipo_bit/ctmi_360/Y	out	HOLD	FALL	min_fall_slack		UNAVAILABLE	NO_FINITE_SLACK

pt_shell

	A	B	C	D	E	F	G	H	I	J	K	L
1	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/sipo_slice_first/ctmi_362	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/A	in	SETUP	RISE	max_rise_slack		UNAVAILABLE
2	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/A	in	SETUP	RISE	max_rise_slack		UNAVAILABLE
3	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/A	in	SETUP	FALL	max_fall_slack		UNAVAILABLE
4	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/A	in	HOLD	RISE	min_rise_slack		UNAVAILABLE
5	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/A	in	HOLD	FALL	min_fall_slack		UNAVAILABLE
6	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/VDD	in	SETUP	RISE	max_rise_slack		UNAVAILABLE
7	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/VDD	in	SETUP	FALL	max_fall_slack		UNAVAILABLE
8	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/VDD	in	HOLD	RISE	min_rise_slack		UNAVAILABLE
9	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/VDD	in	HOLD	FALL	min_fall_slack		UNAVAILABLE
10	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/VSS	in	SETUP	RISE	max_rise_slack		UNAVAILABLE
11	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/VSS	in	SETUP	FALL	max_fall_slack		UNAVAILABLE
12	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/VSS	in	HOLD	RISE	min_rise_slack		UNAVAILABLE
13	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/VSS	in	HOLD	FALL	min_fall_slack		UNAVAILABLE
14	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/Y	out	SETUP	RISE	max_rise_slack		UNAVAILABLE
15	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/Y	out	SETUP	FALL	max_fall_slack		UNAVAILABLE
16	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/Y	out	HOLD	RISE	min_rise_slack		UNAVAILABLE
17	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_9/slice_24/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/compile_HFSINV_36721_169252/Y	out	HOLD	FALL	min_fall_slack		UNAVAILABLE
18	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/A	in	SETUP	FALL	max_fall_slack		UNAVAILABLE
19	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/A	in	HOLD	RISE	min_rise_slack	1.145778	MET
20	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/A	in	HOLD	FALL	min_fall_slack	1.118537	MET
21	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/VDD	in	SETUP	RISE	max_rise_slack		UNAVAILABLE
22	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/VDD	in	SETUP	FALL	max_fall_slack		UNAVAILABLE
23	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/VDD	in	HOLD	RISE	min_rise_slack		UNAVAILABLE
24	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/VDD	in	HOLD	FALL	min_fall_slack		UNAVAILABLE
25	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/VSS	in	SETUP	RISE	max_rise_slack		UNAVAILABLE
26	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/VSS	in	SETUP	FALL	max_fall_slack		UNAVAILABLE
27	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/VSS	in	HOLD	RISE	min_rise_slack		UNAVAILABLE
28	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/VSS	in	HOLD	FALL	min_fall_slack		UNAVAILABLE
29	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/Y	out	SETUP	RISE	max_rise_slack		UNAVAILABLE
30	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/Y	out	SETUP	FALL	max_fall_slack		UNAVAILABLE
31	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/Y	out	HOLD	RISE	min_rise_slack	1.118537	MET
32	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/compile_HFSINV_36721_169252	INVX0_RVT	bit_secure_0/sipo_slice_first/ctmi_362/Y	out	HOLD	FALL	min_fall_slack	1.145778	MET
33	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/A	in	SETUP	RISE	max_rise_slack		UNAVAILABLE
34	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/A	in	SETUP	FALL	max_fall_slack		UNAVAILABLE
35	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/A	in	HOLD	RISE	min_rise_slack	1.21721	MET
36	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/A	in	HOLD	FALL	min_fall_slack	1.188203	MET
37	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/VDD	in	SETUP	RISE	max_rise_slack		UNAVAILABLE
38	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/VDD	in	SETUP	FALL	max_fall_slack		UNAVAILABLE
39	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/VDD	in	HOLD	RISE	min_rise_slack		UNAVAILABLE
40	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/VDD	in	HOLD	FALL	min_fall_slack		UNAVAILABLE
41	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/VSS	in	SETUP	RISE	max_rise_slack		UNAVAILABLE
42	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/VSS	in	SETUP	FALL	max_fall_slack		UNAVAILABLE
43	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/VSS	in	HOLD	RISE	min_rise_slack		UNAVAILABLE
44	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/VSS	in	HOLD	FALL	min_fall_slack		UNAVAILABLE
45	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/Y	out	SETUP	RISE	max_rise_slack		UNAVAILABLE
46	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/Y	out	SETUP	FALL	max_fall_slack		UNAVAILABLE
47	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/Y	out	HOLD	RISE	min_rise_slack	1.188203	MET
48	FC	bit_coin.nlib:bit_coin/chip_finish.design	FUNC_0.85V_HOLD	bit_secure_0/slice_4/sipo_bit/ctmi_360	INVX0_RVT	bit_secure_0/sipo_slice_last/ctmi_362/Y	out	HOLD	FALL	min_fall_slack	1.21721	MET

fc_shell

DAY 06 / CORRELATION

WHAT DRIVES TIMING QoR?

FASTER DATA • SHORTER ROUTES • BETTER CLOCK BALANCE

based on these report, we identify the reason of qor improvement are:

QoR IMPROVES

WNS ↑

FAST PATH / LOW DELAY



LOW CELL + NET DELAY

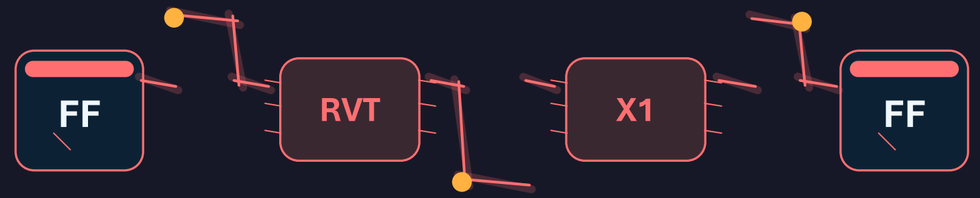


FAST CELLS | SHORT WIRE | USEFUL SKEW | LESS CONGESTION

QoR DEGRADES

WNS ↓

SLOW PATH / HIGH DELAY



HIGH CELL + NET DELAY



SLOW CELLS | LONG WIRE | BAD SKEW | MORE CONGESTION

FC ↔ PT | SAME INPUTS: NETLIST, SDC, PVT, SPEF, SAME PATH

DAY 07
FORMALITY & RTL ECO

DAY 07 • ENGINEERING CHANGE

RTL ECO WORKFLOW

Update a logical function, implement the engineering change and re-verify

01

RTL1

Original RTL

›

02

RTL2

7 AND → OR edits

›

03

GATE ECO

NAND / NOR edits

›

04

FUSION COMPILER

Change mapped cells

›

05

FORMALITY

Check equivalence

WHAT WE ACHIEVED

7 RTL changes identified • Only one (Bit 5 logic updated) •

DAY 07 • ENGINEERING CHANGE

ORIGINAL FORMALITY — PASS

Before ECO: original RTL1 compared with the original gate-level implementation.

```
****************************************** Verification Results ******************************************
Verification SUCCEEDED
   ATTENTION: Verification was run with all UPF supplies constrained to their ON state.
              This is only a partial verification result as it does not cover all
              operational states.
   ATTENTION: synopsys_auto_setup mode was enabled.
              See Synopsys Auto Setup Summary for details.
----------------------------------------------------------------------------------------------
 Reference design: r:/WORK/bit_coin
 Implementation design: i:/WORK/bit_coin
 119990 Passing compare points
----------------------------------------------------------------------------------------------
Matched Compare Points     BBPin    Loop   BBNet     Cut    Port   PDCut     DFF     LAT   TOTAL
----------------------------------------------------------------------------------------------
Passing (equivalent)       28672       0       0    1025     121   10946   79226       0  119990
Failing (not equivalent)       0       0       0       0       0       0       0       0       0
Not Compared
  Don't verify                 0       0       0       0       4       0       0       0       4
**********************************************************************************************
```

VERIFICATION

SUCCEEDED

119,990

Passing compare points

0 failing points

DAY 07 • ENGINEERING CHANGE

FORMALITY & RTL ECO

```
TEST 1: RTL1 vs RTL2_1 (PASS)
-----------------------------
# Linux terminal:
fm_shell

# Formality commands:
read_verilog -r fm_rtl2/rtl1/bitcoin_formality_top.v
set_top r:/WORK/bitcoin_formality_top
read_verilog -i fm_rtl2/rtl2_variants/rtl2_1/bitcoin_formality_top.v
set_top i:/WORK/bitcoin_formality_top
match
verify
exit
```

```
TEST 2: RTL1 vs RTL2_2 (FAIL: bit 0 omitted)
--------------------------------------------
# Linux terminal:
fm_shell

# Formality commands:
read_verilog -r fm_rtl2/rtl1/bitcoin_formality_top.v
set_top r:/WORK/bitcoin_formality_top
read_verilog -i fm_rtl2/rtl2_variants/rtl2_2/bitcoin_formality_top.v
set_top i:/WORK/bitcoin_formality_top
match
verify
report_failing_points
analyze_points -failing
report_analysis_results
exit
```

```
TEST 3: RTL1 vs RTL2_3 (FAIL: XNOR instead of XOR)
--------------------------------------------------
# Linux terminal:
fm_shell

# Formality commands:
read_verilog -r fm_rtl2/rtl1/bitcoin_formality_top.v
set_top r:/WORK/bitcoin_formality_top
read_verilog -i fm_rtl2/rtl2_variants/rtl2_3/bitcoin_formality_top.v
set_top i:/WORK/bitcoin_formality_top
match
verify
report_failing_points
analyze_points -failing
report_analysis_results
exit
```

DAY 07 • ENGINEERING CHANGE

LOGIC PARITY EQUIVALENCE EXERCISE

FORMALITY | BITCOIN RTL EQUIVALENCE RESULTS

Synopsys Formality W-2024.09-SP4 • Reference: rtl1/bitcoin_formality_top.v • 30 Oct 2026

Variant	Prediction*	Tool Result	Matched Inputs	Passing Points	Failing Points	Failing Signal	Root Cause / Explanation
RTL2_1	PASS	PASS	32	1	0	None	Equivalent functionality; implementation source difference not inspected.
RTL2_2	FAIL	FAIL	31	0	1	data_ready	Bit 0 omitted: ^secure_data_out[31:1] instead of ^secure_data_out.
RTL2_3	FAIL	FAIL	32	0	1	data_ready	XNOR (~^) used instead of XOR (^); parity inverted.

SUMMARY	VALUE
Variants tested	3
Passed	1
Failed	2

*Predictions are retrospective, based on source-code inspection; not recorded before running Formality.

Formality FAIL = demonstrated mismatch at data_ready. RTL2_2 bit 0 missing; RTL2_3 inverted parity.

RETROSPECTIVE

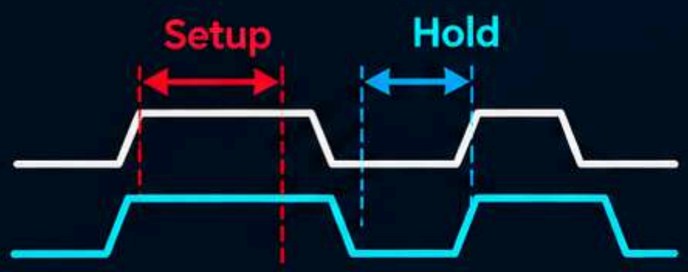
CHALLENGES FACED IN THE BITCOIN PROJECT

Key issues encountered during the physical design implementation

01 TIMING CLOSURE

Challenge

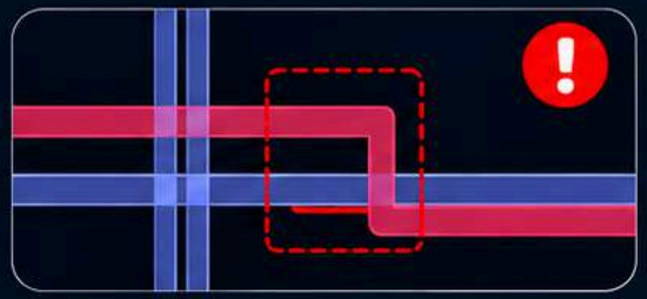
- Meeting timing at the target frequency was challenging.
- Multiple ECO iterations were needed during optimization.
- Had to balance timing, area and power trade-offs.



02 DRC / ROUTING

Challenge

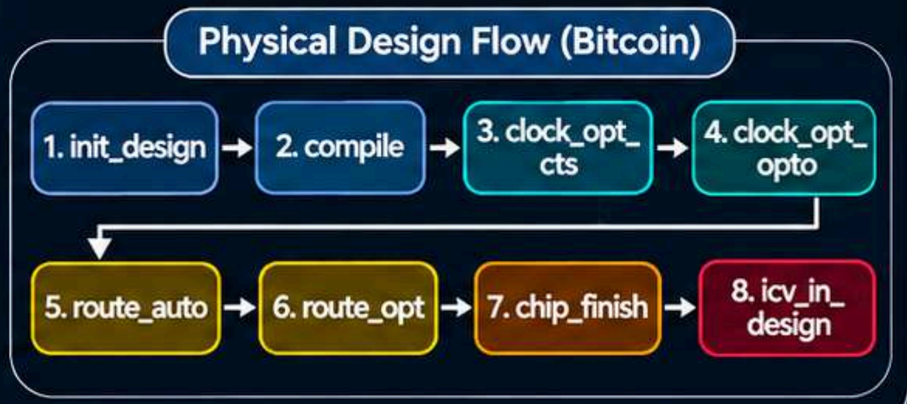
- DRC violations occurred after routing.
- Routing congestion affected some nets and regions.
- Repeated routing cleanup and re-checks were required.



03 PROCESS TIME

Challenge

- The complete flow required a long turnaround time.
- Any iteration or ECO increased the total runtime because multiple stages had to be rerun.



RETROSPECTIVE

WHAT WE LEARNED (OVERALL)

Key takeaways gained from the Bitcoin physical design project



END-TO-END PD UNDERSTANDING

Gained a complete view of the PD flow, from initialization to signoff (ICV).



TIMING & PERFORMANCE

Applied STA and ECO techniques to achieve timing closure at the target frequency.



PPA TRADE-OFF

Understood the trade-off between performance, power and area during optimization.



DESIGN QUALITY & SIGNOFF

Learned to detect, analyze and fix design issues (DRC/LVS) to ensure a clean, signoff-ready design.



ITERATIVE & PRACTICAL MINDSET

Developed problem-solving skills, flow debugging and project management through multiple iterations.



OVERALL TAKEAWAY

Gained practical experience in **physical design implementation**, **timing closure**, **design optimization** (PPA) and **signoff**, using an industry-standard flow and tools.

8-DAY EXECUTION ROADMAP

Bitcoin Physical Design Project – daily themes, leads, and end-of-day deliverables

DAY 1

Intro, planning, fix init_design

Lead: **Syamim**

- Lab copy and 00readme ready
- Roles / daily leads / sync time agreed
- Shared folder, run naming convention and metrics log table created

DAY 2

1G run, study design, key design info

Lead: **Asyraf**

- 1G run finished through chip_finish with no error
- Log-screening summary for errors / warnings
- check_design reports for placement, CTS and route

DAY 3

Analyse 1G, plan frequency sweep

Lead: **Sam**

- 1G timing analysed (WNS, TNS, NVP, worst paths, DRV counts)
- Worst paths mapped to blocks
- Sweep testcase list written and agreed

DAY 4

Pick frequency, tune recipes

Lead: **Anis**

- Sweep results table for all runs
- Target frequency chosen with reason
- Recipe changes listed by stage (compile, placement, CTS, route)

DAY 5

Timing closure + manual ECO

Lead: **Hamzah**

- Best run and target frequency selected
- Remaining setup / hold / max_transition / max_cap violations listed
- Targeted ECO fixes applied with command log

DAY 6

Scripts + FC vs PT correlation

Lead: **Hazim**

- Four analysis scripts validated on Bitcoin results
- Default vs optimized vs final comparison
- FC vs PT comparison for timing, slew, cap, and pin margin

DAY 7

Formality + RTL ECO

Lead: **Asyraf**

- Baseline Formality (RTL1 vs netlist) saved
- RTL1 to RTL2 changes and predicted netlist changes listed
- ECO implemented, placed and routed

DAY 8

Presentation + reports

Lead: **Syamim**

- Final runs frozen and metrics log finalized
- Group report completed and deposited in group_report folder
- 30-minute presentation delivered

Suggested lead rotation shown here. Each day tab contains the full end-of-day checklist.

GROUP 01 / PSDC

THANK YOU

BITCOIN PHYSICAL DESIGN PROJECT

2 NOVEMBER 2026

